



Smart Retail Inventory and Demand Prediction System

Group Name: INFO6007_Activity9_Ji_7

Course Name: INFO6007

AI Usage Declaration

Artificial intelligence tools, including ChatGPT, were used to support the preparation of this Project Management Plan. The use of AI was limited to assisting with brainstorming, structuring sections, improving academic writing, refining grammar, checking consistency across project management documents, and formatting references in Harvard style.

All project decisions, assumptions, project scope, schedule, governance structure, risk responses, quality controls, and final content were reviewed, evaluated, and edited by the group members. The group remained responsible for ensuring that the submitted work accurately reflected the selected project context, course requirements, lecture materials, and relevant academic and industry sources. AI-generated suggestions were not used without human review, and all cited sources were checked and incorporated by the group where appropriate.

Therefore, AI was used as a writing and editing support tool, rather than as a substitute for the group's own analysis, judgement, and project management decision-making.

Contribution Summary

Lihang Shen — 40%

- Completed the Risk Management Plan, Quality Management Plan, Communication Management Plan, Change Management Plan, and Project Monitoring and Control Plan.
- Took responsibility for final integration and consistency checking.
- Check the formatting for the whole paragraph.
- Revised inconsistent or low-quality sections from other members.
- Fixed the Stakeholder Management Plan and created the Power-Interest Grid.

Caresse Zhou — 40%

- Completed Project Scope Management Plan and Project Schedule Management Plan with good quality.
- Revise and improve sections written by other members.
- Check the formatting for the whole paragraph. Helped check consistency with the Project Charter.
- Actively supported the final editing, integration, and quality improvement.

Yiyang Li — 15%

- Provided first drafts of the Project Charter, Stakeholder Management Plan, Literature Review, and Project Governance and Roles. Later revisions were mainly improved by Zhou and Lihang Shen.
- Communication was limited.

Suhani Saha — 5%

- Provided initial drafts of the Project Cost Management Plan and Resource Management Plan with poor quality.
- Missed the internal deadline and rarely responded in WhatsApp.

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Project Charter

Topic	Smart Retail Inventory and Demand Prediction System		
Start Date	26 Feb 2026	End Date	23 June 2026
Brief Background			
The retail industry is currently under pressure from unstable supply chains, changing customer purchasing patterns, and competition from e-commerce platforms. For retailers, poor demand forecasting often leads to two major problems: stockouts of popular products and excessive inventory of slow-moving items. Both issues increase operating costs, reduce profit margins, and affect customer satisfaction. As retail operations become more time-sensitive and data-intensive, traditional rule-based inventory management systems are becoming less effective. The target organisation is a mid-sized Australian retail chain with 45 store locations. It currently manages inventory through a reactive, rule-based process supported by separate ERP, WMS, and POS systems. Because these systems are not fully connected, the organisation has limited visibility across sales, warehouse stock, and procurement activities. This makes it difficult to respond accurately to changes in demand. As a result, the organisation experiences a bullwhip effect, where small changes in customer demand can create larger disruptions in ordering and stock control. This has contributed to repeated stockouts of fast-moving consumer goods during peak periods, as well as the build-up of dead stock in lower-demand categories. To address these issues, this project will develop and deploy a Smart Retail Inventory and Demand Prediction System. The system will use machine learning models, including LSTM and XGBoost, to generate SKU-level demand forecasts and support automated procurement decisions. It will also connect with the organisation's existing ERP, WMS, and POS systems so that sales, inventory, and purchasing information can be used in a more coordinated way. This will help the organisation move from relying mainly on past reports to making more timely and forward-looking inventory decisions. The project is expected to improve inventory turnover, product availability, and operational efficiency within four months of commencement. It also supports the organisation's broader digital transformation goals by improving how inventory decisions are made across stores, warehouses, and procurement teams. By reducing stockouts and excess inventory, the system will help the organisation protect profit margins and remain competitive in the Australian retail market.			
Project Objectives			
• Design and deploy a cloud-native Artificial Intelligence (AI) forecasting engine utilising LSTM and XGBoost frameworks to predict SKU-level demand. • Establish secure, bi-directional API middleware to enable real-time inventory synchronisation across existing legacy POS, WMS, and ERP systems. • Develop a deterministic backend rules engine to automatically generate ERP-compliant Purchase Orders based on dynamic safety stock calculations. • Build a centralised, web-based operational dashboard featuring real-time inventory mapping and a Human-in-the-Loop (HITL) procurement console. • Execute a comprehensive change management and training programme for all 45 Store Managers and central procurement staff prior to system deployment.			
Project Success Criteria			
• Algorithm Performance: Achieve a baseline demand forecast accuracy of $\geq 85\%$ (measured via MAPE) at the Store-SKU level within 90 days of Go-Live. • Operational Efficiency: Attain a verifiable reduction in out-of-stock (OOS) incidents by $\geq 30\%$ compared to the pre-project baseline after 6 months.			

- **Financial Impact:** Decrease excess inventory holding costs by $\geq 20\%$ within the 6-month post-deployment review period.
- **System Reliability:** Guarantee an API integration data loss rate of exactly zero, with an overall system uptime of $\geq 99.5\%$ post-deployment.
- **Organisational Adoption:** Secure a system usage and adoption rate of $\geq 90\%$ across the targeted operational staff within 60 days of launch.

Project Milestones

Milestone 1 — Project Initiation & Charter Approved: 04 March 2026
 Milestone 2 — Cloud & Data Infrastructure Established: 25 March 2026
 Milestone 3 — AI Model & API Middleware Complete: 29 April 2026
 Milestone 4 — System Integration Testing (SIT) Sign-off: 20 May 2026
 Milestone 5 — User Acceptance Testing (UAT) Sign-off: 03 June 2026
 Milestone 6 — System Go-Live & Staff Trained: 10 June 2026
 Milestone 7 — Project Closure (End of Hypercare): 23 June 2026

Budget Information

Total Approved Budget: AUD 500,000

- System Design & Architecture: AUD 60,000 (12%)
- AI/ML Model Development: AUD 120,000 (24%)
- System Integration & Development: AUD 100,000 (20%)
- Cloud Infrastructure & Hosting: AUD 80,000 (16%) (Note: Strict cap as per Scope baseline)
- Testing & Quality Assurance: AUD 50,000 (10%)
- Project Management, Training & Change Mgmt: AUD 40,000 (8%)
- Contingency Reserve: AUD 50,000 (10%)

Roles and Responsibilities

Name	Responsibilities	Role
Lihang Shen	Lead overall project planning, scheduling, budgeting, stakeholder communication, and risk management. Chair weekly team meetings and prepare progress reports.	Project Manager
Yiyang Li	Conduct requirements gathering, document business processes, coordinate UAT with end users, and manage stakeholder feedback.	Business Analyst
Qianyou Zhou	Design system architecture, develop AI/ML demand forecasting models, oversee technical integrations with ERP/POS/WMS systems.	Technical Lead / ML Engineer
Suhani Saha	Develop and execute test plans, perform QA, manage user training sessions, and support change management activities.	QA & Change Management Lead

Signature of Stakeholders

Lihang Shen
 Yiyang Li
 Qianyou Zhou
 Suhani Saha

Stakeholder Management Plan

Introduction

Stakeholder management is an important part of delivering the Smart Retail Inventory and Demand Prediction System because the project involves many groups with different responsibilities and expectations. These include executive sponsors, the IT department, inventory and procurement staff, store managers, external vendors, and system integration partners. If these stakeholders are not clearly identified and managed, the project may face delays, unclear requirements, resistance from users, or disagreements over scope and priorities.

For this project, stakeholder management is used to ensure that the people and groups affected by the system are involved at the right time and through the right communication channels. Early engagement is particularly important because the new system will change how inventory decisions are made, how procurement activities are triggered, and how store-level stock information is monitored. Understanding stakeholder expectations at the beginning of the project will help reduce resistance, improve requirement accuracy, and support smoother adoption during deployment. Previous research also highlights that active stakeholder engagement is closely linked to IT project success (Eslerod, Huemann and Savage, 2015).

This Stakeholder Management Plan explains how stakeholders will be identified, categorised, analysed, and engaged throughout the project lifecycle. It covers the period from project initiation and requirements gathering through to deployment and post-go-live review. The plan is informed by stakeholder management research, including Jepsen and Eslerod (2009), and will be reviewed at each major project milestone to ensure that stakeholder needs, concerns, and levels of influence remain properly understood.

Stakeholder Identification

Stakeholders for this project were identified through a structured approach combining project charter review, organisational chart analysis, expert interviews with department heads, and facilitated brainstorming sessions with the core project team. This multi-method approach ensures comprehensive coverage of all parties who may affect or be affected by the project.

Internal Stakeholders are individuals and teams within the organisation directly involved in, or impacted by, the implementation of the system. They include the Project Sponsor (Head of Digital Transformation), the core project team (Project Manager, Business Analyst, Technical Lead, and QA Lead), the IT Department responsible for infrastructure, the Inventory and Procurement Team as primary end-users, Store Managers who will interact with the system daily, the Finance Department overseeing budget compliance, and the Data Analytics Team supporting model validation.

External Stakeholders include parties outside the organisation whose involvement is essential to project success. These include retail suppliers and vendors who will interface with the automated replenishment engine, the cloud services provider (e.g., AWS or Azure), the ML solution vendor, the ERP software vendor, and independent system integration consultants. Regulatory bodies overseeing data privacy compliance (e.g., Australian Privacy Act) are also considered key external stakeholders.

Internal Stakeholders	External Stakeholders
1. Project Sponsor (Head of Digital Transformation)	1. Retail Suppliers / Vendors 2. Cloud Services Provider (AWS/Azure)

2. Project Manager 3. IT Department 4. Inventory & Procurement Team 5. Store Managers 6. Data Analytics Team 7. Finance Department	3. ML Solution Vendor 4. ERP Software Vendor 5. System Integration Consultants 6. Regulatory Bodies (Australian Privacy Act) 7. End Customers (indirect beneficiaries)
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Stakeholder Analysis

Each stakeholder has been assessed across three dimensions: power (ability to influence project decisions), interest (degree of concern in project outcomes), and impact (extent to which the project affects their work or objectives).

The Project Sponsor holds high power and high interest — their continued support is essential for securing budget approvals and resolving escalated issues. The IT Department and Technical Lead possess high power due to their control over infrastructure and technical delivery, and high interest as direct contributors. Store Managers and the Inventory Team hold moderate power but very high interest as primary end-users; resistance from this group could jeopardise adoption.

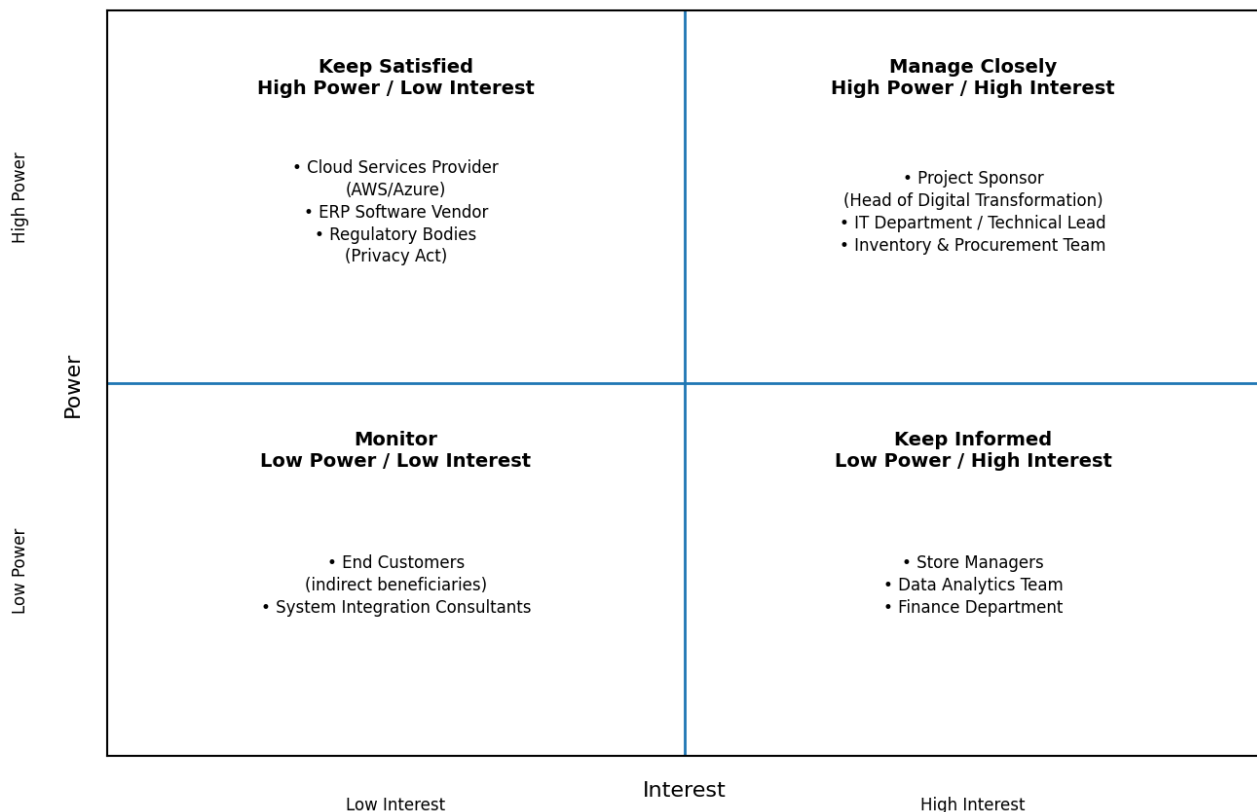
External stakeholders such as the cloud services provider and ERP vendor hold high power (as technical dependencies) but lower interest in organisational outcomes. Retail suppliers have moderate interest and moderate power, particularly in the context of automated purchase order generation. Regulatory bodies have low day-to-day interest but high latent power, as non-compliance could halt the project.

Potential areas of resistance include store managers concerned about operational disruption during transition, and procurement staff worried about role changes resulting from automation. These must be managed through targeted communication and change management strategies.

Power Interest Grid

The Power-Interest Grid below categorises stakeholders into four quadrants to guide engagement strategies:

Power-Interest Grid for Stakeholder Engagement



HIGH POWER / HIGH INTEREST — Manage Closely:

- Project Sponsor (Head of Digital Transformation)
- IT Department / Technical Lead
- Inventory & Procurement Team

HIGH POWER / LOW INTEREST — Keep Satisfied:

- Cloud Services Provider (AWS/Azure)
- ERP Software Vendor
- Regulatory Bodies (Privacy Act)

LOW POWER / HIGH INTEREST — Keep Informed:

- Store Managers
- Data Analytics Team
- Finance Department

LOW POWER / LOW INTEREST — Monitor:

- End Customers (indirect beneficiaries)
- System Integration Consultants

Engagement Strategies

Engagement strategies have been tailored to each stakeholder group based on their position in the Power-Interest Grid:

- **Manage Closely (High Power, High Interest):** The Project Sponsor will receive weekly written status reports and participate in bi-weekly steering committee meetings. The IT Department and

core team will engage through daily stand-up meetings, shared project management tools (e.g., Jira/Confluence), and a dedicated project communication channel (Microsoft Teams). The Inventory and Procurement Team will be involved in requirements workshops and UAT sessions.

- **Keep Satisfied (High Power, Low Interest):** Vendors (cloud provider, ERP vendor) will be managed via formal contract reviews, SLA dashboards, and monthly performance meetings. Regulatory bodies will be engaged through compliance reports and legal counsel reviews at key project gates.
- **Keep Informed (Low Power, High Interest):** Store Managers will receive fortnightly project newsletters and participate in dedicated change management workshops before go-live. The Finance Department will receive monthly budget snapshots. The Data Analytics Team will be consulted during model validation phases.
- **Monitor (Low Power, Low Interest):** End customers and integration consultants will be monitored passively, with ad-hoc communication as needed. No formal engagement cadence is required for this group at this stage.

Communication Linkage

This Stakeholder Management Plan is directly linked to the Communication Management Plan, which provides the detailed framework for all project communications. The communication preferences and frequency defined above (e.g., steering committee meetings, newsletters, dashboards, workshops) are formally documented and governed within the Communication Management Plan.

All stakeholder communication will adhere to the information needs, formats, and escalation paths defined in that plan, ensuring consistency, traceability, and alignment across the project lifecycle. Any changes to stakeholder engagement approaches will be reviewed in accordance with the Change Management Plan and reflected in both documents.

Literature Review

Introduction

This literature review supports the development of a comprehensive Project Management Plan (PMP) for the Smart Retail Inventory and Demand Prediction System. Its purpose is to ground the plan in established theory and contemporary practice by synthesising relevant academic and industry literature. The review examines three interconnected areas: project management frameworks applicable to complex IT initiatives, the specific management challenges posed by AI and machine learning (ML) technologies, and identified gaps in current practice that directly inform decisions made throughout this PMP. Insights drawn from the literature are used to justify methodological choices across governance, scope, schedule, risk, and stakeholder management sections.

Literature and Industry Context

Project management of large-scale IT initiatives remains a persistent challenge for organisations globally. Dwivedi et al. (2015) conducted a comprehensive review of IS project failure and success research, confirming that the majority of IT initiatives continue to fall short of their original scope, time, or budget targets, a pattern documented consistently across industries and geographies. These findings underscore the importance of applying rigorous, evidence-based project management frameworks.

Gemino, Reich and Serrador (2021) examined 477 cross-industry projects and found that hybrid project management approaches, combining the structured planning of traditional methodologies with the adaptive capacity of Agile, achieved significantly higher stakeholder satisfaction than purely traditional approaches, while delivering comparable outcomes in terms of time, budget, and scope. Their findings directly inform the methodology adopted in this PMP: a hybrid model that accommodates milestone-based governance to satisfy executive stakeholders, while permitting iterative ML model development and testing cycles.

Governance and stakeholder management emerge consistently in the literature as primary determinants of IT project success. Müller and Turner (2007) found that project governance structures, including clearly defined roles, decision rights, and escalation pathways, significantly reduce the likelihood of scope creep and budget overrun. Flyvbjerg and Budzier (2011) further demonstrated in a study of 1,471 IT projects that one in six exceeded budgets by an average of 200%, often due to inadequate risk governance. These findings directly motivate the robust governance structure, RACI matrix, and risk management processes embedded in this PMP.

Furthermore, Fernandez and Fernandez (2008) argue that hybrid models are particularly suited to enterprise IT projects where requirements are partially defined upfront but subject to change during development, a characteristic that closely mirrors this retail AI initiative.

Regarding technology-specific delivery outcomes, Feizabadi (2022) demonstrated statistically significant supply chain performance improvements from machine learning-based demand forecasting over traditional methods, while also noting that data quality and integration complexity are the primary barriers to operationalising such systems in practice. This finding directly informs the project's emphasis on ERP/POS integration planning and structured change management activities.

Technology Considerations for Project Management

The Smart Retail Inventory and Demand Prediction System relies on several interdependent technologies, including machine learning forecasting models (e.g., Long Short-Term Memory networks and gradient boosting algorithms such as XGBoost), real-time data pipelines, cloud-based infrastructure, and API-based integrations with existing ERP and POS systems. While the technical design is addressed separately, these technology choices carry significant project management implications across four key dimensions.

Scope and Complexity: ML model development is inherently experimental. Unlike conventional software development, ML projects require iterative data collection, feature engineering, model training, and validation cycles before a production-ready model can be defined. Lwakatare et al. (2019) identify unique technical debt patterns and scope uncertainty in ML systems that must be explicitly accounted for in the Work Breakdown Structure and scope change procedures, necessitating a clearly defined scope boundary that separates model development iterations from the overall project timeline.

Risk and Uncertainty: AI projects introduce non-deterministic risk: model performance depends on data quality, volume, and representativeness, which may not be fully known at project initiation. Amershi et al. (2019) document that data-related challenges are the leading cause of AI project delays in enterprise settings. Accordingly, data readiness assessment has been incorporated as a formal early milestone in the project schedule.

Resource and Skill Requirements: The project requires specialised ML engineering expertise distinct from traditional IT project skills. Chuang et al. (2024) demonstrate that ML and AI technology creates significant skill gaps in organisations, with roles such as ML engineers and AI data scientists requiring continuous development pathways that many organisations have not yet established. This creates resourcing risks that must be addressed in the Resource Management Plan.

Schedule and Cost: Cloud infrastructure costs for ML training workloads are variable and difficult to estimate accurately upfront. GPU-accelerated compute instances on cloud platforms introduce cost uncertainty that requires contingency budgeting and close monitoring throughout execution.

Knowledge Gaps and Project Implications

Despite growing literature on AI project management, several gaps remain relevant to this initiative. First, most existing research on AI demand forecasting focuses on technical performance metrics rather than the project management processes required to deliver such systems in complex organisational environments (Lwakatare et al., 2019). There is limited guidance on how to structure governance for hybrid AI/IT projects where development cycles differ substantially from traditional software delivery.

Second, while change management for ERP implementations is well-studied (Somers and Nelson, 2004), the literature on managing user adoption specifically for AI-driven retail tools remains nascent. The behavioural resistance from operational staff — who may distrust algorithmic recommendations — represents a challenge for which standardised mitigation frameworks are underdeveloped.

These gaps inform two design decisions in this PMP: (1) the adoption of a hybrid governance model that accommodates both structured delivery milestones and iterative ML development sprints; and (2) the inclusion of a dedicated change management stream within the project, with specific activities targeting store manager and procurement team adoption.

Summary

The literature reviewed here establishes a clear evidence base for the project management decisions embedded in this PMP. Hybrid Agile-waterfall delivery methods, supported by Gemino et al. (2021), provide the methodological foundation. Lessons from comparable ML-based supply chain deployments (Feizabadi, 2022) highlight the critical importance of data integration planning and change management, both of which have been explicitly incorporated as workstreams. The governance structure and RACI matrix respond to empirical evidence linking clear decision rights to IT project success (Müller and Turner, 2007). Technology-specific risks, particularly data readiness (Amershi et al., 2019), ML skill gaps (Chuang et al., 2024), and variable cloud costs, are addressed in the Risk and Resource Management Plans respectively. Together, these insights ensure that the project is grounded in both theoretical rigour and practical precedent.

Project Governance and Roles

Introduction

Project governance in IT project management refers to the framework of policies, processes, roles, and decision-making structures that ensure a project is delivered in an accountable, controlled, and transparent manner. Effective governance aligns project execution with organisational strategy, ensures resources are used efficiently, and provides clear mechanisms for oversight, risk escalation, and issue resolution.

For the Smart Retail Inventory and Demand Prediction System, robust governance is particularly critical given the project's technical complexity, cross-functional dependencies, and significant investment. Sirisomboonsuk et al. (2018) surveyed 533 IT professionals and found that both project governance and IT governance have a statistically significant positive impact on project performance — specifically across portfolio direction, project sponsorship, and disclosure and reporting dimensions. Without clearly defined accountability and decision rights, IT projects of this scale are prone to scope creep, communication failures, and delayed delivery (Müller and Turner, 2007). This section defines the project's governance structure, the roles and responsibilities of all key participants, a RACI matrix for major project activities, and the processes through which decisions will be made, documented, and communicated throughout the project lifecycle.

Governance Structure

The project will operate under a three-tier governance structure:

Tier 1 — Executive Oversight (Project Sponsor): The Project Sponsor (Head of Digital Transformation) provides strategic direction, final approval on scope and budget changes exceeding AUD \$20,000, and escalation resolution. The Sponsor chairs a monthly Steering Committee meeting attended by the Project Manager and key departmental heads.

Tier 2 — Project Management (Project Manager): The Project Manager is responsible for day-to-day delivery, progress reporting, risk and issue management, and coordinating across all functional workstreams. The PM facilitates weekly team stand-up meetings and produces fortnightly status reports for the Sponsor.

Tier 3 — Delivery Team (Technical Lead, Business Analyst, QA Lead): The delivery team executes all technical and analytical work, reports status to the Project Manager, and escalates blockers through defined channels. Daily stand-ups are held within the technical team during active development sprints.

This hierarchy ensures clear lines of accountability, fast decision-making at the operational level, and appropriate executive oversight for high-impact decisions. Decision escalation is triggered when issues cannot be resolved within 48 hours at the delivery team level.

Roles and Responsibilities

Project Sponsor (Head of Digital Transformation): Provides executive sponsorship, approves the project charter and budget, champions the initiative at the board level, and makes final decisions on significant scope or budget changes. Accountable for ensuring the project aligns with organisational

strategy.

Project Manager: Responsible for overall planning, schedule management, budget tracking, risk and issue management, stakeholder communication, and project reporting. The single point of accountability for delivery against the PMP.

Business Analyst: Leads requirements gathering, documents user stories and functional specifications, facilitates workshops with end-user groups, and coordinates User Acceptance Testing (UAT). Ensures the technical solution meets business needs.

Technical Lead / ML Engineer: Responsible for system architecture design, ML model development (demand forecasting algorithms), API integrations with ERP/POS/WMS systems, and cloud infrastructure setup. Technical authority for all build and integration decisions.

QA & Change Management Lead: Develops and executes the test strategy (unit, integration, UAT), manages defect tracking, coordinates user training programmes, and implements the change management plan to support adoption. Ensures quality gates are met before each phase transition.

IT Department: Provides infrastructure support, network access, security compliance review, and ongoing technical environment management. Consulted on architectural decisions with IT policy implications.

Inventory & Procurement Team: Primary end-users of the system. Participates in requirements validation, UAT, and training. Provides subject matter expertise on current inventory workflows and replenishment processes.

RACI Matrix

The RACI matrix below defines accountability for all major project activities across the key project roles. R = Responsible (does the work), A = Accountable (owns the outcome), C = Consulted (provides input), I = Informed (receives updates). Each activity has exactly one Accountable owner.

Activity	Project Sponsor	Project Manager	Business Analyst	Technical Lead
Project Planning & Scheduling	A	R	C	C
Requirements Definition	I	A	R	C
ML Model Development	I	A	C	R
System Integration (ERP/POS/WMS)	I	A	C	R
Risk Management	I	R	C	C
Change Control Approval	A	R	C	C
Testing & UAT	I	A	C	C
User Training	I	A	C	I
Deployment & Go-Live	A	R	I	C
Budget Management	A	R	I	I
Stakeholder Reporting	I	R	I	I

Governance Process

All project decisions are classified into three tiers based on their impact on scope, cost, or schedule. Khalid et al. (2022) empirically demonstrated that inadequate project governance and IT governance are leading antecedents of schedule delay — reinforcing the need for clearly defined decision thresholds and escalation pathways.

Tier 1 — Operational Decisions (Project Manager authority): Routine decisions with no impact on approved baseline. Examples include task assignment changes, minor schedule adjustments within a sprint, and day-to-day resource allocation. Documented in the project log within 24 hours.

Tier 2 — Tactical Decisions (PM + Technical Lead): Changes that affect technical design or schedule within a single workstream, without exceeding a cost variation of AUD \$20,000. Requires PM written approval and is documented in the change log and communicated to the Sponsor.

Tier 3 — Strategic Decisions (Project Sponsor approval required): Any change that affects the approved baseline by more than AUD \$20,000, alters project scope, or shifts the go-live date by more than two weeks. A formal Change Request must be submitted, reviewed at the next Steering Committee meeting, and approved by the Sponsor before implementation.

All decisions are recorded in the project decision register, with rationale, owner, and date. Issues that cannot be resolved within 48 hours are escalated to the next governance tier.

Compliance

Project governance ensures compliance across three domains:

Organisational Policies: The project will adhere to the organisation's IT governance policies, procurement procedures, and data management standards. All vendor contracts will be reviewed by the Legal and Procurement teams prior to signing, and technology choices will be reviewed for alignment with the enterprise architecture roadmap. The governance framework adopted in this project aligns with the three-dimension model of project governance: portfolio direction, project sponsorship, and disclosure and reporting, validated by Sirisomboonsuk et al. (2018) as positively associated with project performance outcomes.

Data Privacy and Security: The system will handle sensitive retail transaction data. Compliance with the Australian Privacy Act 1988 (Cth) and the Australian Privacy Principles (APPs) is mandatory. A Privacy Impact Assessment (PIA) will be conducted prior to system go-live, and data encryption, access controls, and audit logging will be implemented in accordance with the organisation's information security policy. Available at: <https://www.legislation.gov.au/Details/C2022C00235> (Accessed: 16 April 2026).

Quality and Standards: The project will embed quality gate reviews at each major milestone to assess delivery against functional, reliability, usability, and performance criteria. Governance checkpoints will be used to formally verify that each phase has met its defined acceptance criteria before proceeding. Khalid et al. (2022) confirm that integrating governance controls at phase transitions significantly reduces the likelihood of schedule and quality deviations in complex IT projects.

Project Scope Management Plan

Project	Smart Retail Inventory and Demand Prediction System		
Start Date	26 Feb 2026	Prepared by	Caresse Zhou

Justification	The modern retail environment is characterised by macro-economic supply chain volatility, omnichannel complexity, and rapid shifts in consumer demand. The target organisation (a mid-sized Australian retail chain operating 45 locations) currently relies on reactive, rule-based inventory management and siloed legacy infrastructures (ERP, WMS, and POS systems). This fragmented approach creates a severe "bullwhip effect," leading to two critical financial pain points: chronic stockouts of fast-moving consumer goods (FMCG) during peak demand, and the accumulation of dead stock that locks up working capital and degrades overall profit margins. To secure its market position, the organisation requires a paradigm shift from retrospective reporting to predictive, autonomous operations. The Smart Retail Inventory and Demand Prediction System justifies its capital investment by leveraging enterprise-grade Machine Learning (e.g., gradient boosting frameworks like XGBoost and time-series neural networks like LSTM) to process high-dimensional historical transactional data. By operationalising these AI models via a scalable, secure cloud architecture (e.g., AWS/Azure) and establishing real-time API integrations with existing core systems, the project will transform inventory procurement from a manual, heuristic-driven process into a highly precise, algorithmic capability. This initiative is the cornerstone of the organisation's digital transformation roadmap, designed to deliver verifiable Return on Investment (ROI) through optimised inventory turnover, minimised holding costs, and maximised product availability.
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Objectives	The project objectives are defined across five fundamental pillars of delivery. These objectives adhere to SMART criteria and serve as the binding acceptance criteria for project closure and vendor sign-off. 1. AI/ML Modeling & MLOps Objectives 1.1 Forecast Accuracy Target: Design, train, and deploy a cloud-native Machine Learning forecasting engine capable of achieving a baseline forecast accuracy of $\geq 85\%$ at the Store-SKU level (measured via Mean Absolute Percentage Error - MAPE) within 90 days of system go-live. 1.2 Automated Retraining Pipeline: Implement an automated MLOps pipeline (e.g., AWS SageMaker pipelines) that triggers model retraining bi-weekly using the latest POS data to prevent data drift and adapt to shifting seasonal trends.
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1.3 Cold-Start & Edge-Case Handling: Develop explicit algorithmic heuristics to handle "cold-start" scenarios for newly introduced SKUs and highly sparse sales data items, ensuring 100% of the active catalogue receives a baseline forecast.

2. System Integration & Data Engineering Objectives

2.1 API Connectivity: Establish secure, bi-directional RESTful API endpoints between the new AI prediction engine and the legacy Enterprise Resource Planning (ERP), Point of Sale (POS), and Warehouse Management Systems (WMS).

2.2 Data Integrity Guarantee: Ensure zero critical data loss, duplication, or corruption during the historical data migration phase and ongoing real-time operational syncs.

2.3 Latency & Performance SLAs: Maintain an API response latency of <500ms for real-time inventory delta syncs during peak operational retail hours (9:00 AM – 6:00 PM).

2.4 Security & Compliance: Enforce end-to-end data encryption (AES-256 at rest, TLS 1.3 in transit) and implement role-based access control (RBAC) in strict compliance with the Australian Privacy Act 1988.

3. Application Frontend & Operational Dashboard Objectives

3.1 Enterprise UI Deployment: Develop and deploy a centralised, responsive Web User Interface (UI) accessible via standard enterprise web browsers for the 45 Store Managers and the central Procurement Team.

3.2 Real-Time Inventory Mapping: Deliver a dashboard module providing real-time visualisation of current stock levels across the central distribution centre and all physical retail nodes.

3.3 Predictive Alerting System: Implement a configurable push-notification system to flag projected stockouts or overstock risks with a minimum 7-day predictive lead time.

3.4 Human-in-the-Loop (HITL) Console: Build an override interface allowing senior procurement staff to review, manually modify, or approve ML-generated restock quantities prior to final execution.

4. Automated Replenishment & Execution Objectives

4.1 Dynamic Purchase Order Generation: Implement a deterministic rules-engine that consumes ML demand predictions to automatically generate formatted Purchase Orders (POs) compliant with existing ERP standards.

	4.2 Dynamic Safety Stock Calculation: Configure the engine to dynamically calculate and continuously adjust safety stock thresholds based on specific vendor lead times and predicted demand volatility. 4.3 Process Efficiency Gain: Reduce the manual processing time required for routine procurement and PO generation by a minimum of 80% compared to current operational baselines. 5. Business Value, Reliability & Adoption Objectives 5.1 Stockout Reduction: Attain a verifiable reduction in out-of-stock (OOS) incidents by $\geq 30\%$ against the FY2025 pre-project baseline, evaluated at the 6-month post-deployment milestone. 5.2 Holding Cost Reduction: Decrease excess inventory holding costs by $\geq 20\%$ within the same 6-month post-deployment review period. 5.3 High Availability SLA: Guarantee enterprise-grade system reliability with a measured Uptime of $\geq 99.5\%$, excluding explicitly approved and scheduled maintenance windows over the first three months. 5.4 Target User Adoption: Secure a system user adoption rate of $\geq 90\%$ across all target users within 60 days of launch, driven by a structured Change Management programme and mandatory training workshops.
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Summary of Project Deliverables	
PM Deliverables	Cloud Data Ingestion & ETL Pipeline - Functional Scope: Automated scripts responsible for extracting, transforming, and loading (ETL) historical transaction, promotion, and inventory data (from January 1, 2023, onwards) from legacy ERP and POS systems into a central cloud data lake. - Delivery Format & Platform: Python-based ETL scripts deployed via a cloud-native orchestration service (e.g., AWS Glue or Apache Airflow). Delivered as version-controlled source code (via enterprise GitHub/GitLab) along with executable deployment templates. - Application Scenario: Executes autonomously as a scheduled nightly batch job (e.g., at 2:00 AM) during non-peak retail hours to update the central repository without impacting daytime POS terminal performance. AI Demand Forecasting Module - Functional Scope: The core machine learning engine utilizing LSTM and XGBoost algorithms to predict SKU-level demand across all 45 store locations, achieving $\geq 85\%$ accuracy (MAPE). - Delivery Format & Platform: Serialised model artifacts (e.g., .pkl or ONNX formats) packaged within lightweight Docker containers. Deployed to a managed machine learning cloud

	environment (e.g., Amazon SageMaker) exposing secure inference endpoints. ● Application Scenario: Polled dynamically by the central dashboard and replenishment engine every morning to generate a rolling 7-day forecast for the active catalogue of 15,000 SKUs. Automated MLOps Retraining Pipeline ● Functional Scope: A continuous integration/continuous deployment (CI/CD) pipeline designed to automatically retrain the forecasting models using the latest transactional data to prevent data drift. ● Delivery Format & Platform: Infrastructure-as-Code (IaC) configuration files (e.g., Terraform or AWS CloudFormation YAML files) and pipeline scripts. ● Application Scenario: Runs as a background service on the 1st and 15th of every month. It trains a "challenger" model against the new data and automatically routes production traffic to the new model only if it mathematically outperforms the baseline "champion" model. Bi-Directional API Integration Middleware ● Functional Scope: The critical communication layer that translates and routes data between the new cloud AI system and the legacy on-premise ERP (e.g., SAP/Microsoft Dynamics), WMS, and POS networks with zero critical data loss. ● Delivery Format & Platform: A suite of RESTful APIs documented in OpenAPI 3.0 (Swagger) format, deployed as serverless functions (e.g., AWS Lambda) behind a secure API Gateway. ● Application Scenario: Acts as a real-time bridge during business hours, ensuring that when a physical item is scanned at a POS terminal, the delta is securely transmitted (TLS 1.3 encrypted) to the cloud system within a latency threshold of <500ms. ● Automated Purchase Order (PO) Execution Engine ● Functional Scope: A backend business rules engine that consumes the AI forecasts, applies vendor lead-time constraints and dynamic safety stock logic, and calculates replenishment quantities. ● Delivery Format & Platform: Backend microservice application. The output delivery format of the POs will be strictly formatted data files (e.g., JSON, XML, or EDIFACT) that natively match the ingestion requirements of the existing legacy ERP system. ● Application Scenario: Triggered daily at 4:00 PM, the engine automatically consolidates the day's predictions and pushes formatted draft POs directly into the ERP's procurement module, effectively eliminating manual data entry. Web-Based Operational Dashboard & UI
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	● Functional Scope: The centralised user interface featuring a real-time inventory map, predictive 7-day stockout alerts, and a Human-in-the-Loop (HITL) console for PO review and approval. ● Delivery Format & Platform: A responsive Single Page Application (SPA) developed in React or Angular. Platform explicitly defined: This is a web-only application accessible via standard enterprise desktop browsers (Chrome 110+, Edge 110+) and responsive for enterprise-managed tablet browsers (iPadOS Safari). Note: Native mobile applications (iOS App Store / Android Google Play APKs) are strictly out of scope. ● Application Scenario: Used daily by the 45 Store Managers via tablets while walking the retail floor to check stock statuses, and used concurrently by the central Procurement Team via desktop monitors for bulk PO approvals. Provisioned & Secured Cloud Infrastructure ● Functional Scope: The foundational hosting environment required to run all aforementioned modules securely, guaranteeing $\geq 99.5\%$ uptime while adhering to the AUD \$80,000 budget cap. ● Delivery Format & Platform: A fully operational enterprise cloud environment (e.g., AWS or Azure) encompassing Virtual Private Clouds (VPCs), Load Balancers, Identity and Access Management (IAM) roles, and GPU compute instances. Delivered with a comprehensive Architectural Topology Diagram (e.g., Visio/Lucidchart). ● Application Scenario: Serves as the 24/7 production environment for the system, configured with AES-256 data-at-rest encryption to ensure strict compliance with the Australian Privacy Act 1988 for all processed retail data.
Product Deliverables	Comprehensive Project Management Plan (PMP) & Subsidiary Plans ● Functional Scope: The master governance artifact that consolidates all planning dimensions of the project, including Scope, Schedule, Cost, Resource, Risk, Quality, Communication, and Change Management Plans. It establishes the formal baselines against which project performance will be measured. ● Delivery Format & Platform: A formally formatted, consolidated PDF document conforming to academic and industry standards (e.g., PMBOK). Delivered via the organisation's enterprise document management system (e.g., SharePoint/Confluence). ● Application Scenario: Serves as the binding execution contract. It requires a formal digital signature from the Project Sponsor (Head of Digital Transformation) prior to the commencement of the Execution phase. Requirements Traceability Matrix (RTM) ● Functional Scope: A core quality assurance tool that bi-directionally maps every approved business requirement from the

	Inventory & Procurement Team to its corresponding technical architecture design, ML model feature, and final test script. • Delivery Format & Platform: A dynamic, multi-tab enterprise spreadsheet (e.g., Microsoft Excel) or a configured requirements tracking board in Jira. • Application Scenario: Utilised by the QA Lead during System Integration Testing (SIT) and User Acceptance Testing (UAT) to mathematically prove that 100% of the contracted features have been built and rigorously tested. Live Governance Dashboards & Change Control Logs • Functional Scope: The continuous tracking mechanisms for project health, encompassing the Risk Register, Issue Log, and the formal Change Control Log for recording all Tier 2 and Tier 3 strategic decisions. • Delivery Format & Platform: Hosted live on a collaborative project management platform (e.g., Jira/Confluence). Change Control approvals will be documented as formal IT service tickets with timestamped audit trails. • Application Scenario: Actively projected and reviewed by the Project Manager during weekly delivery team stand-ups and formally presented at the monthly Steering Committee meetings with the Project Sponsor. Testing Strategy & UAT Sign-Off Certificates • Functional Scope: Detailed documentation defining the testing methodology for the AI forecasting accuracy (targeting $\geq 85\%$), system integration (API latency $< 500\text{ms}$), and user workflows. It includes the final, legally binding sign-off sheets confirming the system is ready for production. • Delivery Format & Platform: Formal testing procedure manuals (PDF) and digital sign-off certificates routed through an e-signature platform (e.g., DocuSign). • Application Scenario: Executed at the conclusion of Milestone 5. The UAT certificates must be physically or digitally signed by the representative Subject Matter Experts from the Inventory Team and Store Managers to authorize the final Go-Live phase. Change Management & User Training Asset Library • Functional Scope: Educational and change-enablement materials specifically engineered to secure the target $\geq 90\%$ user adoption rate among operational staff, addressing potential behavioral resistance. • Delivery Format & Platform: A comprehensive digital package including: two digital "Quick Reference Guides" (QRGs) in PDF format, interactive training modules (SCORM compliant for the corporate LMS), and presentation slide decks. • Application Scenario: Deployed directly to the 45 Store Managers and central Procurement Team during the scheduled
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	half-day training workshops in mid-April 2026, acting as both teaching aids and long-term reference manuals. Go-Live Cutover Runbook & Post-Implementation Review (PIR) • Functional Scope: The Runbook details the minute-by-minute technical deployment steps, IT support rotas, and emergency rollback procedures for the system launch. The PIR is the final closure document evaluating overall project success, budget variance, and lessons learned. • Delivery Format & Platform: The Runbook is delivered as an actionable sequence spreadsheet. The PIR is delivered as a formal management report (Word/PDF) and an executive presentation deck. • Application Scenario: The Runbook is strictly followed by the Technical Lead and IT Department during the deployment weekend (Milestone 6). The PIR is presented by the Project Manager to the executive board at the formal Project Closure meeting.
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Project Scope	
In Scope	This section defines the strict, legally binding boundaries of the functional deliverables and operational services to be executed by the project team. Any activity, service, or feature not explicitly listed below is excluded from the project baseline. 1. Cloud Infrastructure & Security Provisioning Design, configuration, and provisioning of three distinct cloud environments (Development, UAT, and Production) on an enterprise cloud platform (e.g., AWS or Azure) using Infrastructure-as-Code (IaC), strictly capped at the allocated AUD \$80,000 budget. Configuration of network security, including Virtual Private Clouds (VPCs) and API firewalls, applying AES-256 (data-at-rest) and TLS 1.3 (data-in-transit) encryption to ensure explicit compliance with the Australian Privacy Act 1988. 2. Data Engineering & AI Model Development Development of automated ETL pipelines to extract, clean, and map historical transactional and inventory data sourced exclusively from January 1, 2023, to the present date. Engineering and training of machine learning algorithms (LSTM and XGBoost) to predict demand for the active catalogue of approximately 15,000 SKUs, targeting a baseline accuracy of $\geq 85\%$.

	Model Tuning Rework Limit: Algorithmic tuning is strictly limited to three (3) major hyperparameter optimization cycles during the development sprint to reach the 85% accuracy threshold. Development of an automated MLOps pipeline for bi-weekly model retraining in production. 3. Application & Middleware Engineering Development of a RESTful API gateway to serve as the bi-directional middleware enabling real-time data syncs (<500ms latency) between the cloud AI engine and the legacy on-premise ERP, WMS, and POS systems. Development of a web-based, responsive operational dashboard featuring a real-time inventory map, predictive stockout alerts, and a Human-in-the-Loop (HITL) procurement console. Development of an automated replenishment rules engine that computes safety stock and formats output data to match the native Purchase Order (PO) ingestion format of the existing ERP. UI/UX Rework Limit: The development of the web dashboard includes a maximum of two (2) rounds of client feedback and interface revision prior to the formal "Design Freeze" sign-off. 4. Testing, UAT Acceptance & Defect Rework Policy Execution of System Integration Testing (SIT) by the Technical Lead to verify API stability and data integrity with zero critical data loss. Facilitation of a strictly time-boxed fourteen (14) day User Acceptance Testing (UAT) window for the Inventory & Procurement Team and selected Store Managers. Acceptance & Rework Protocol: During UAT, the project team is responsible for remediating all identified Severity 1 (Critical System Failure) and Severity 2 (Core Functionality Impaired) defects. The scope includes a maximum of two (2) formal UAT bug-fix iteration deployment cycles. Scope Protection: Any user dissatisfaction arising from missing features that were not documented in the baseline Requirements Traceability Matrix (RTM) will not be treated as a "defect." Such requests will be rejected as Out of Scope and must be routed through the formal Tier 3 Change Request process for separate budget and schedule assessment. 5. Training, Go-Live, & Post-Deployment Support (Hypercare)
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	Delivery of four (4) targeted, half-day functional training workshops for the 45 Store Managers and central procurement staff, alongside the provision of digital Quick Reference Guides (QRGs). Execution of the technical Go-Live cutover over a single weekend (Milestone 6). Hypercare Period (Post-Deployment Maintenance): Provision of a post-deployment Hypercare support period from 11 June 2026 to 23 June 2026 immediately following System Go-Live. During this window, the core project team will resolve any latent Severity 1 or 2 system bugs and monitor ML model stability at no additional project cost. System Handover: Delivery of technical architecture documentation, runbooks, and a formal knowledge transfer session to the internal IT Department to assume ongoing Tier 1 and Tier 2 operational support after the 30-day Hypercare period concludes.
Out of Scope	This section explicitly identifies the products, services, processes, and support mechanisms that are strictly excluded from the project baseline. Any request to incorporate the following items will be immediately rejected or must be processed as a formal Tier 3 Change Request requiring additional budget and timeline extensions. Hardware & Physical Infrastructure Procurement - The procurement, installation, maintenance, or configuration of any physical hardware. This explicitly excludes purchasing or upgrading Point-of-Sale (POS) terminals, warehouse barcode scanners, networking cables, routers, or on-premise physical servers for the 45 retail locations or central distribution centres. Legacy System Code Modification & Upgrades - Refactoring, patching, or modifying the internal core codebase, database schemas, or proprietary business logic of the legacy on-premise ERP, WMS, or POS platforms. The project team's responsibility is strictly limited to mapping to and calling the existing vendor-provided APIs. Any system upgrades required on the legacy side to support API connectivity are the sole responsibility of the organisation's IT Department or the respective software vendors. Historical Data Cleansing & Backfilling - Manual auditing, cleansing, data-entry correction, or back-filling of corrupted, missing, or unstructured historical data generated prior to January 1, 2023. The project assumes that the data extracted via existing systems from 2023 onwards is structurally valid. If historical data is missing or highly corrupted, the AI model will not mathematically interpolate it. Native Mobile Apps & Consumer-Facing Interfaces

	• The design or development of any native mobile applications (e.g., iOS App Store / Android Google Play APKs). The operational dashboard is explicitly defined as a web-based responsive application. Furthermore, the development of any e-commerce portals, customer loyalty apps, or direct-to-consumer tracking interfaces is strictly excluded. Extended System Support & Maintenance (Beyond Hypercare) • The provision of Tier 1, Tier 2, or Tier 3 IT helpdesk support, ongoing server maintenance, or manual machine learning model recalibration beyond the explicitly defined post-deployment Hypercare period ending on 23 June 2026. Following this Hypercare window, full operational ownership and maintenance accountability transfer completely to the organisation's internal IT Department. Cosmetic Rework & Out-of-SLA Defect Resolution • Remediation of Severity 3 (Minor) or Severity 4 (Cosmetic) defects during the User Acceptance Testing (UAT) phase if they do not actively impede core business functionalities or algorithm accuracy. Furthermore, any UI/UX modification requests submitted after the formal "Design Freeze" milestone, or exceeding the two (2) allotted revision cycles defined in the scope, will not be executed. Financial Processing & Logistics Execution • Automated financial clearing, vendor payment processing, or invoice settlement. While the automated engine will generate draft Purchase Orders (POs), the actual financial transaction and approval workflows remain entirely within the legacy ERP's finance modules. Additionally, physical supply chain optimisation, such as warehouse layout redesign or third-party freight routing, is excluded. Ongoing Cloud Consumption & Vendor Licensing Fees • The financial payment or ongoing assumption of third-party software licensing fees, ERP vendor API integration surcharges, or recurring cloud platform consumption costs (e.g., AWS monthly billing) occurring after the project's formal Go-Live and Closure dates. The AUD \$80,000 cloud budget applies exclusively to the development, testing, and deployment phases
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Project Assumptions

This section documents the foundational assumptions upon which the project's scope, schedule, and cost baselines have been constructed. Should any of these baseline assumptions be proven false during project execution, it will constitute a valid trigger for the Project Manager to initiate the formal Tier 3 Change Request process to reassess the schedule and budget.

1. Legacy System API Readiness & Vendor Cooperation

It is assumed that the organisation's existing legacy ERP (e.g., SAP/Microsoft Dynamics), WMS, and POS software possess standard, documented API connectivity (RESTful, GraphQL, or standard EDI/XML export capabilities) without requiring bespoke middleware to be built from scratch on the legacy side. Furthermore, it is assumed that the legacy software vendors will provide access to accurate API documentation and testing sandbox environments within fourteen (14) days of project kick-off.

2. Historical Data Quality & Structural Integrity

It is assumed that the historical transactional, promotional, and inventory data generated from January 1, 2023, onwards is structurally sound, continuous, and representative of actual retail operations. The project schedule and budget assume that the ML models can achieve the targeted $\geq 85\%$ accuracy without necessitating a massive, unbudgeted manual data-cleansing or back-filling phase to correct corrupt legacy records.

3. Algorithmic Feasibility & Demand Predictability

It is assumed that the consumer purchasing behaviour across the 15,000 SKUs exhibits mathematically identifiable patterns (seasonality, trend, or cyclicity). The targeted $\geq 85\%$ SKU-level forecast accuracy is assumed to be inherently achievable via industry-standard ML algorithms (LSTM, XGBoost) based on the quality of the provided POS data.

4. Cloud Cost Stability & Resource Provisioning

It is assumed that the pricing models for the selected enterprise cloud provider (e.g., AWS or Azure) will remain stable throughout the project lifecycle. Specifically, the variable costs associated with on-demand GPU-accelerated compute instances required for iterative Machine Learning model training will not experience unpredictable price surges and will be fully accommodated within the AUD \$80,000 cloud infrastructure budget.

5. Store Network Infrastructure & Bandwidth

It is assumed that all 45 physical retail store locations and the central distribution centres currently possess stable, commercial-grade internet infrastructure. The existing network bandwidth is assumed to be sufficient to support real-time POS data synchronisation to the central cloud platform without exceeding the $< 500\text{ms}$ latency threshold during peak retail operating hours.

6. Stakeholder Engagement & SME Availability

It is assumed that all identified Key Stakeholders, specifically the primary Subject Matter Experts (SMEs) from the Inventory & Procurement Team and selected Store Managers, will be made available to participate in scheduled requirements gathering workshops, User Acceptance Testing (UAT), and training sessions. It is assumed they will allocate a minimum of 6 hours per week during critical phase-gate milestones without reprioritising their operational duties over project commitments.

7. Executive Enforcement & Adoption Mandate

It is assumed that the Project Sponsor (Head of Digital Transformation) possesses the organisational authority and executive willingness to mandate the use of the new system. It is assumed that executive management will actively enforce the system's adoption to mitigate potential behavioural resistance from operational staff, ensuring the $\geq 90\%$ adoption target can be met within 60 days.

8. Regulatory & Compliance Stability

It is assumed that there will be no immediate, retroactive legislative amendments to the Australian Privacy Act 1988 (Cth) or Australian Data Sovereignty regulations during the 4-month project execution window (Feb - June 2026) that would necessitate a mid-development architectural redesign of the data encryption or cloud storage protocols.

9. Core Team Retention & Skill Adequacy

It is assumed that the designated core project delivery team (Project Manager, Business Analyst, Technical Lead/ML Engineer, and QA/Change Management Lead) will remain intact for the entirety of the project lifecycle. It is also assumed that the existing enterprise IT Department possesses the baseline technical competency required to assume Tier 1 and Tier 2 support handover upon the conclusion of the 30-day Post-Deployment Hypercare period.

Project Schedule Management Plan

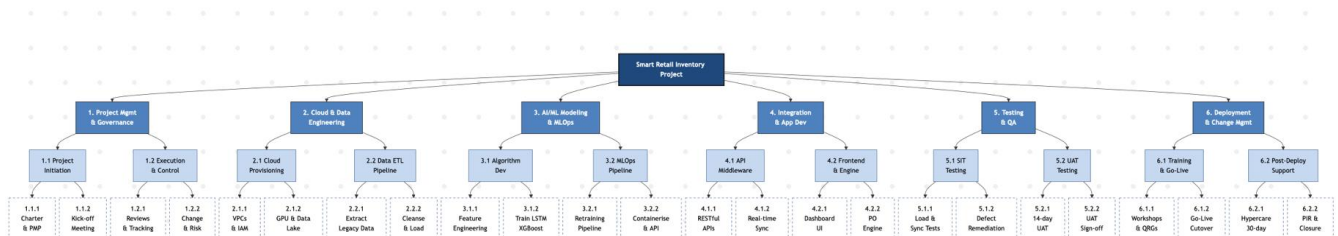
Introduction

Schedule management in complex enterprise IT projects, particularly those involving predictive Artificial Intelligence (AI) and legacy system integrations, is the definitive process of planning, developing, managing, and controlling the project timeline to ensure punctual delivery. In the context of the Smart Retail Inventory and Demand Prediction System, a rigorous Schedule Management Plan is critical to address the inherent uncertainties of machine learning model training and to mitigate schedule risks associated with cross-functional dependencies, such as API readiness from legacy ERP vendors. This plan establishes a definitive temporal baseline to coordinate the efforts of the technical delivery team, align executive stakeholder expectations, and guarantee that the system achieves its operational go-live target without compromising testing rigor or data integrity.

By adopting a hybrid scheduling methodology that utilises rigid Waterfall milestones for executive governance alongside iterative Agile sprints for ML algorithm development, this document outlines the explicit processes for developing and controlling the project schedule. It details the Work Breakdown Structure (WBS), maps sequential task dependencies via Network Diagramming, applies the Program Evaluation and Review Technique (PERT) for probabilistic duration estimation, and identifies the core Critical Path. Ultimately, this framework ensures that all project activities remain synchronised, technical resource allocations are optimised, and potential schedule variances are proactively managed.

Schedule Development Process

The Work Breakdown Structure

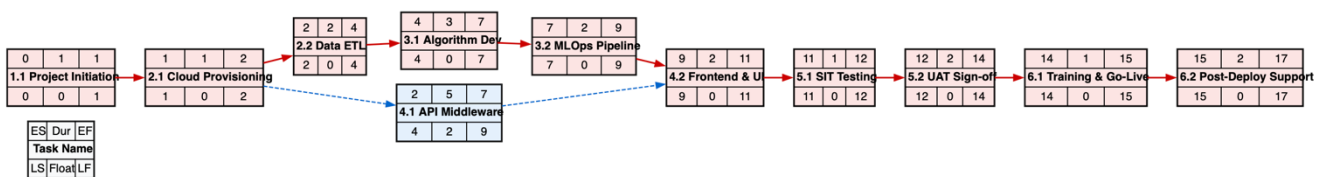


WBS Item No	Description	Dependencies (if any)
1	Project Management & Governance	
1.1	Project Initiation: Establish project baseline, team, and initial documentation.	
1.1.1	Draft and approve Project Charter and Project Management Plan (PMP).	None
1.1.2	Conduct formal Project Kick-off meeting with the Steering Committee.	1.1.1
1.2	Project Execution & Control: Ongoing tracking of schedule, budget, and risks.	
1.2.1	Facilitate weekly stand-ups and monthly Steering Committee reviews.	1.1.2

1.2.2	Manage Tier 2/Tier 3 Change Requests and update the Risk Register.	1.1.2
2	Cloud Infrastructure & Data Engineering	
2.1	Cloud Environment Provisioning: Set up AWS/Azure resources.	
2.1.1	Configure Virtual Private Clouds (VPCs), IAM roles, and firewalls.	1.1.2
2.1.2	Provision GPU-accelerated compute instances and data lake storage.	2.1.1
2.2	Data ETL Pipeline Development: Build historical data ingestion scripts.	
2.2.1	Extract historical POS, WMS, and ERP data (Jan 2023 onwards).	2.1.2
2.2.2	Cleanse, transform, and load data into the secure cloud data lake.	2.2.1
3	AI/ML Modeling & MLOps	
3.1	Algorithm Development: Train the core demand forecasting models.	
3.1.1	Perform feature engineering on ingested historical retail data.	2.2.2
3.1.2	Train and tune LSTM and XGBoost models to achieve $\geq 85\%$ accuracy.	3.1.1
3.2	MLOps Pipeline Integration: Automate the model lifecycle.	
3.2.1	Develop and deploy the bi-weekly automated model retraining pipeline.	3.1.2
3.2.2	Containerise approved models (e.g., Docker) and expose inference endpoints.	3.1.2
4	System Integration & Application Development	
4.1	API Middleware Development: Build the bi-directional communication bridge.	
4.1.1	Develop RESTful APIs connecting the AI engine to legacy ERP/WMS/POS.	2.1.1
4.1.2	Configure real-time data syncs ensuring <500ms latency and TLS 1.3 encryption.	4.1.1
4.2	Frontend & Replenishment Engine: Build the UI and business logic.	
4.2.1	Develop the Web-Based Operational Dashboard (inventory map, alerts, HITL).	4.1.1
4.2.2	Code the automated rules engine to generate formatted Purchase Orders.	3.2.2, 4.1.1
5	Testing & Quality Assurance	
5.1	System Integration Testing (SIT): Validate technical architecture.	
5.1.1	Execute API load testing and verify zero critical data loss during syncs.	4.1.2, 4.2.2
5.1.2	Remediate any Severity 1 or 2 defects identified during SIT.	5.1.1
5.2	User Acceptance Testing (UAT): Validate business requirements.	
5.2.1	Facilitate 14-day UAT with the Procurement Team and Store Managers.	5.1.2
5.2.2	Secure formal UAT sign-off certificates from key stakeholders.	5.2.1
6	Deployment & Change Management	

6.1	Training & Go-Live: Execute user enablement and system launch.	
6.1.1	Deliver half-day training workshops and distribute Quick Reference Guides.	5.2.2
6.1.2	Execute Go-Live cutover runbook and deploy system to production.	6.1.1
6.2	Post-Deployment Support: Project closure and handover.	
6.2.1	Provide 15-days Hypercare support to resolve latent production bugs.	6.1.2
6.2.2	Deliver Post-Implementation Review (PIR) and formally close project.	6.2.1

Network Diagram



The Network Diagram illustrates the logical sequencing and interdependencies of all Level 2 project activities using the Precedence Diagramming Method (PDM). The critical path—representing the longest sequence of dependent tasks with zero float—dictates the shortest

possible project duration of 17 weeks. A strategic architectural decision was made to execute Track A (Data & AI/ML Modeling) and Track B (API Middleware Development) concurrently. Because the API middleware development requires less time than the highly iterative ML modeling phase, it inherently possesses two weeks of schedule float. This deliberate parallel routing optimises resource allocation, allowing the Technical Lead to shift focus between the two tracks without jeopardising the final integration point at the Application Development phase.

Estimating resources for Activities

WBS Item No	Resources
1.1	Human: Project Sponsor, Project Manager. Software: Confluence, MS Word, DocuSign (for charter approval).
1.2	Human: Project Manager. Software: Jira (for tracking), MS Teams (communications), MS Project (scheduling).
2.1	Human: Technical Lead, IT Department. Technical/Hardware: AWS/Azure Cloud Subscriptions, VPC, IAM, GPU Compute Instances. Software: Terraform (Infrastructure-as-Code).
2.2	Human: Technical Lead/ML Engineer, Data Analytics Team.

	Technical: Cloud Data Lake Storage (e.g., AWS S3). Software: Python, Apache Airflow / AWS Glue.
3.1	Human: Technical Lead/ML Engineer, Data Analytics Team. Technical: GPU Compute Instances (for heavy model training). Software: Jupyter Notebooks, XGBoost/LSTM libraries, Python.
3.2	Human: Technical Lead/ML Engineer. Technical: Cloud ML Environment (e.g., AWS SageMaker). Software: Docker, GitHub/GitLab (for CI/CD version control).
4.1	Human: Technical Lead/ML Engineer, IT Department. Technical: API Gateway Serverless compute. Software: Swagger (OpenAPI 3.0), Postman (for API dev).
4.2	Human: Business Analyst, Technical Lead/ML Engineer. Software: React/Angular (Frontend UI), Node.js/Python (Backend Engine), IDEs (VS Code).
5.1	Human: QA Lead, Technical Lead/ML Engineer. Software: Selenium (UI testing), JMeter (API load testing), Jira (Defect tracking).
5.2	Human: QA Lead, Business Analyst, Inventory Team SMEs, Store Managers. Technical: UAT Cloud Environment (Staging). Software: Jira (Defect logging), DocuSign (UAT Sign-off).
6.1	Human: QA & Change Management Lead, Project Manager, Store Managers. Software: Corporate LMS (Learning Management System), PDF Readers (for QRGs), Production Cloud Environment.

Estimating durations

To establish a realistic and mathematically sound project timeline, the duration for each Level 2 activity was calculated using the Program Evaluation and Review Technique (PERT). Because machine learning development and legacy system integration carry inherent uncertainties, PERT provides a weighted average duration rather than a single-point estimate.

The duration in weeks (E) was calculated using the standard PERT formula:

$$E = \frac{O + 4M + P}{6}$$

(Where O = Optimistic estimate, M = Most likely estimate, and P = Pessimistic estimate).

For example, for WBS 3.1 (Algorithm Development), the estimates were: Optimistic (2 weeks), Most Likely (3 weeks), Pessimistic (4 weeks). Using the formula, the expected duration resolves to exactly 3 weeks.

Note: WBS 1.2 (Project Execution & Control) is a Level of Effort (LOE) activity that spans the entire lifecycle of the project, hence its duration encompasses the full 17 weeks.

WBS Item No.	Start Date	End Date	Duration (weeks)
1.1	26 Feb 2026	04 Mar 2026	1
1.2	26 Feb 2026	23 Jun 2026	17
2.1	05 Mar 2026	11 Mar 2026	1
2.2	12 Mar 2026	25 Mar 2026	2
3.1	26 Mar 2026	15 Apr 2026	3
3.2	16 Apr 2026	29 Apr 2026	2
4.1	12 Mar 2026	15 Apr 2026	5
4.2	30 Apr 2026	13 May 2026	2
5.1	14 May 2026	20 May 2026	1
5.2	21 May 2026	03 Jun 2026	2
6.1	04 Jun 2026	10 Jun 2026	1
6.2	11 Jun 2026	23 Jun 2026	2

Critical Path Method

Based on the precedence logic defined in the Network Diagram and the PERT duration estimates, the Critical Path Method (CPM) was applied to identify the longest continuous sequence of activities from project initiation to project closure. This path determines the shortest possible time in which the project can be completed (17 weeks). Any delay in a critical activity will directly delay the final Go-Live and project closure dates.

1. The Critical Path The mathematical Critical Path for this project is formally identified as: **Start → 1.1 → 2.1 → 2.2 → 3.1 → 3.2 → 4.2 → 5.1 → 5.2 → 6.1 → 6.2 → End**

2. Critical Activities (Zero Slack / Zero Float) The activities listed below lie directly on the critical path. They have zero weeks of total float/slack, meaning their early start/finish times equal their late start/finish times. Close monitoring of these tasks is paramount:

- 1.1 Project Initiation
- 2.1 Cloud Environment Provisioning
- 2.2 Data ETL Pipeline Development
- 3.1 Algorithm Development
- 3.2 MLOps Pipeline Integration
- 4.2 Frontend & Replenishment Engine
- 5.1 System Integration Testing (SIT)
- 5.2 User Acceptance Testing (UAT)

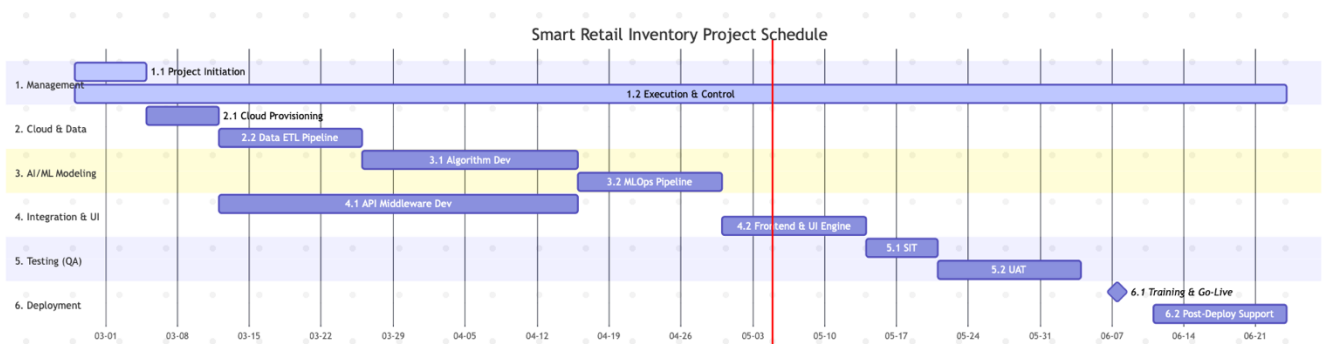
- 6.1 Training & Go-Live
- 6.2 Post-Deployment Support

3. Non-Critical Activities and Slack/Float Analysis To optimise resource utilisation, **WBS 4.1 (API Middleware Development)** is deliberately scheduled to run in parallel with the data and machine learning tracks (WBS 2.2, 3.1, and 3.2).

- **Path A (Data/ML Track - Critical):** WBS 2.2 (2w) + 3.1 (3w) + 3.2 (2w) = **7 weeks total duration.**
- **Path B (API Track - Non-Critical):** WBS 4.1 = **5 weeks total duration.**
- **Float/Slack Identification:** Since Path A and Path B converge before WBS 4.2 can begin, **WBS 4.1 possesses a Total Slack/Float of exactly two (2) weeks.** This strategic buffer means that if unexpected complexities arise while integrating with the legacy ERP/POS APIs, the middleware development team has a 2-week safety margin before their delays begin to impact the overall project go-live date.

(Note: WBS 1.2 Project Execution & Control is an overarching support activity with a fixed duration bound to the project timeline, and thus carries zero float.)

Gantt Chart



The Gantt Chart translates the network logic into a chronological project timeline, spanning from project initiation on 26 February 2026 to formal closure on 23 June 2026. This schedule arrangement is deliberately structured using a Hybrid Agile-Waterfall methodology to mitigate specific technology delivery risks. Sequential, fixed Waterfall milestones are enforced for the initial Cloud Provisioning and final UAT/Go-Live phases to ensure strict executive governance, security compliance, and operational stability.

Conversely, the AI/ML Modeling, API Integration, and Frontend UI phases are purposefully arranged to overlap. This schedule compression technique, known as fast-tracking, allows the engineering team to conduct iterative Agile development sprints. By testing API endpoints and tuning ML models simultaneously, the team avoids waiting for one distinct phase to entirely finish before starting the next.

Furthermore, the schedule deliberately sets the System Go-Live to occur by early June. This strategy intentionally reserves the final segment of the project timeline exclusively for a dedicated Post-Deployment Support (Hypercare) window, ensuring any latent production defects are resolved prior to the absolute project closure deadline.

Review of the Schedule

To ensure the viability, accuracy, and operational feasibility of the project timeline, a comprehensive, multi-step schedule review was conducted prior to baselining the schedule. The review process actively involved the Project Manager, Technical Lead, and key stakeholders to validate the logical flow of activities against the constrained 17-week timeframe (26 Feb 2026 – 23 Jun 2026). The specific actions taken to review and validate the schedule were as follows:

1. **WBS Completeness & Scope Alignment Check:** The schedule was cross-referenced against the Scope Management Plan and the WBS Dictionary. We verified the 100% rule to ensure that all Level 2 and Level 3 deliverables were fully represented as distinct schedule activities with no missing work packages. This verification specifically included the critical machine learning model training, API integrations, and UAT phases.
2. **Dependency & Network Logic Verification:** A forward and backward pass analysis was conducted on the Precedence Diagramming Method (PDM) network diagram. This confirmed that all Finish-to-Start (FS) dependencies were logical. Specifically, we verified that the convergence point (WBS 4.2 Frontend & Replenishment Engine) correctly waits for both Track A (MLOps) and Track B (API Middleware) to finish before UI integration can commence.
3. **Resource Feasibility & Leveling Analysis:** We cross-checked the *Estimating resources for Activities* matrix against the parallel timeline. Because the Technical Lead is required on both the Data/ML track and the API Middleware track simultaneously, we verified that the calculated 2-week float/slack on the API track allows for adequate "Resource Leveling." This ensures the Technical Lead is not mathematically overallocated (>100% capacity) during weeks 3 to 7.
4. **PERT Estimate Validation:** The Optimistic, Pessimistic, and Most Likely duration estimates used in the PERT calculations were reviewed with the Technical Lead. We specifically ensured that sufficient buffer was built into WBS 3.1 (Algorithm Development) to account for the iterative nature of ML hyperparameter tuning.
5. **Milestone & Objective Alignment Check:** Finally, the calculated Critical Path dates were mapped against the strategic business objectives. We confirmed that the scheduled Go-Live (10 June 2026) safely precedes the final project closure deadline (23 June 2026), leaving a robust 2-week window exclusively for the required Post-Deployment Hypercare support.

Milestones

The following table details the major project milestones, marking the completion of critical phases and the delivery of key phase-gate artifacts. These dates are derived directly from the Critical Path Method calculation and serve as the primary progress tracking points for the Steering Committee.

Sr No.	Milestone Name	Description	Date of achievement
1	Project Initiation & Charter Approved	Formal approval of the Project Management Plan (PMP) and Project Charter by the Project Sponsor. Marks the end of WBS 1.1.	04 Mar 2026
2	Cloud & Data Infrastructure Established	Successful provisioning of the AWS/Azure environment and completion of the	25 Mar 2026

		automated ETL data ingestion pipeline (WBS 2.2).	
3	AI Model & API Middleware Complete	Convergence of parallel development tracks. ML models are trained ($\geq 85\%$ accuracy) and legacy API middleware is fully developed (WBS 3.2 & 4.1).	29 Apr 2026
4	System Integration Testing (SIT) Sign-off	Frontend dashboard, automated PO engine, and backend models are integrated. Technical Lead signs off on zero critical data loss and $< 500\text{ms}$ API latency (WBS 5.1).	20 May 2026
5	User Acceptance Testing (UAT) Sign-off	Completion of the 14-day testing window. The Inventory & Procurement Team formally signs UAT certificates, authorizing production deployment (WBS 5.2).	03 Jun 2026
6	System Go-Live & Staff Trained	Successful execution of the Go-Live cutover runbook over the weekend. Target user base has completed all necessary change management training (WBS 6.1).	10 Jun 2026
7	Project Closure & Handover	Conclusion of the post-deployment Hypercare window. Delivery of the Post-Implementation Review (PIR) and formal handover to IT support. (WBS 6.2).	23 Jun 2026

Cost Management Plan

Introduction

Project cost management is the process of planning, estimating, budgeting, financing, monitoring, and controlling project costs to ensure that the project can be completed within the approved budget. For IT projects, effective cost management is particularly important because project expenditure is often affected by specialised human resources, cloud infrastructure, software tools, vendor dependencies, system integration complexity, cybersecurity requirements, and testing activities.

For the Smart Retail Inventory and Demand Prediction System, cost management is essential because the project involves multiple high-cost workstreams, including AI/ML model development, cloud infrastructure provisioning, API middleware development, ERP/POS/WMS integration, system testing, user training, and change management. Without a structured cost management process, the project may experience budget overruns, uncontrolled cloud consumption, unexpected integration costs, or insufficient funding for testing and deployment.

This Cost Management Plan defines how project costs will be estimated, approved, allocated, monitored, and controlled throughout the project lifecycle. The total approved budget, as stated in the Project Charter, is **AUD 500,000**. This budget includes all planned project expenditure and contingency reserve. The plan therefore establishes the cost baseline against which actual spending will be measured. It also defines how funding will be released, how cost variance will be reviewed, and how budget changes will be managed through the project governance and change control process.

Cost estimation approach

The project will use a bottom-up cost estimation approach. This means that the cost of each major workstream will be estimated separately and then consolidated into the total project budget. This approach is suitable for the Smart Retail Inventory and Demand Prediction System because the project contains several distinct cost areas, including system design, AI/ML development, integration, cloud infrastructure, testing, project management, training, change management, and contingency reserve.

The cost estimates are based on the approved project scope and the budget categories defined in the Project Charter. The key assumptions used in the estimation process are that the project will run from **26 February 2026 to 23 June 2026**, the organisation will use existing ERP, POS, and WMS systems, the project team will not replace existing legacy platforms, and the system will be hosted on an enterprise cloud platform such as AWS or Azure. It is also assumed that the existing retail data from 2023 onwards is structurally usable for AI/ML model training, and that the organisation's internal IT Department and external vendors will provide required technical support during integration and testing.

Several constraints are considered in the cost estimation. First, the total project budget is fixed at **AUD 500,000**, so all project costs must remain within this limit. Second, the cloud infrastructure and hosting budget is capped at **AUD 80,000**, which requires active monitoring of GPU compute, storage, API hosting, and backup costs. Third, the implementation timeline is short, so cost planning must ensure that sufficient resources are available during the most intensive development, integration, and testing phases. Fourth, any unplanned scope expansion, additional model tuning, vendor delay, or API integration complexity may create additional cost pressure.

The Project Manager, Finance Department, Technical Lead, QA and Change Management Lead, and relevant vendor representatives will review cost estimates before approval. During execution, cost control will be supported through cost baseline management, variance analysis, Earned Value Management, cost performance monitoring, contingency reserve management, and formal change control procedures.

Cost Breakdown

Type of Cost: System Design & Architecture

Detail	Unit	Hours	Cost (AUD)	Total (AUD)
Technical Lead / System Architect	1	180	150	27,000
Business Analyst	1	150	100	15,000
Architecture design tools and documentation	1	-	8,000	8,000
Architecture review workshops and vendor consultation	1	-	10,000	10,000
Total				60,000

Type of Cost: AI/ML Model Development

Detail	Unit	Hours	Cost (AUD)	Total (AUD)
ML Engineers	2	300	130	78,000
Data Analytics Support	1	160	100	16,000
GPU training and experimentation environment	1	-	16,000	16,000
Model validation and technical documentation	1	-	10,000	10,000
Total				120,000

Type of Cost: System Integration & Development

Detail	Unit	Hours	Cost (AUD)	Total (AUD)
Software Developers	2	300	110	66,000

API integration tools and middleware services	1	-	12,000	12,000
Automated replenishment engine development support	1	-	12,000	12,000
Integration documentation and code review	1	-	10,000	10,000
Total				100,000

Type of Cost: Cloud Infrastructure & Hosting

Detail	Unit	Hours	Cost (AUD)	Total (AUD)
Cloud Engineer	1	180	100	18,000
Cloud hosting, compute, and storage services	1	-	50,000	50,000
Backup, recovery, and monitoring services	1	-	12,000	12,000
Total				80,000

Type of Cost: Testing & Quality Assurance

Detail	Unit	Hours	Cost (AUD)	Total (AUD)
Quality Assurance / Testing Team	2	180	80	28,800
UAT coordination support	1	80	90	7,200
Testing tools and test environment setup	1	-	8,000	8,000
Defect tracking and quality reporting	1	-	6,000	6,000
Total				50,000

Type of Cost: Project Management, Training & Change Management

Detail	Unit	Hours	Cost (AUD)	Total (AUD)
Project Manager	1	220	120	26,400

Change Management / Training Lead	1	120	70	8,400
Training materials, Quick Reference Guides, and LMS resources	1	-	5,200	5,200
Total				40,000

Type of Cost: Contingency Reserve

Detail	Unit	Hours	Cost (AUD)	Total (AUD)
Contingency reserve for technical, integration, cloud, testing, and deployment risks	-	-	50,000	50,000
Total				50,000

Cost Baselines

The project cost baseline represents the approved time-phased budget across the project lifecycle. It will be used to monitor planned expenditure against actual expenditure and to support cost variance analysis. Since the project runs for approximately four months, the baseline is divided into four monthly periods.

WBS / Cost Area	Phase 1 Initiation & Planning	Phase 2 Design & Development	Phase 3 Integration & Testing	Phase 4 Deployment & Closure	Total (AUD)
System Design & Architecture	30,000	25,000	5,000	-	60,000
AI/ML Model Development	-	90,000	30,000	-	120,000
System Integration & Development	-	40,000	50,000	10,000	100,000
Cloud Infrastructure & Hosting	10,000	30,000	30,000	10,000	80,000

Testing & Quality Assurance	-	-	40,000	10,000	50,000
Project Management, Training & Change Management	10,000	10,000	5,000	15,000	40,000
Contingency Reserve	10,000	15,000	15,000	10,000	50,000
Total	50,000	210,000	175,000	65,000	500,000

The cost baseline is aligned with the project schedule. Month 1 focuses on initiation, planning, system design, early cloud setup, and project management. Month 2 has the highest planned expenditure because it includes major AI/ML development, system design completion, cloud infrastructure usage, and early system integration work. Month 3 includes continued integration, model validation, testing, and quality assurance activities. Month 4 covers final testing, deployment, training, go-live support, closure activities, and remaining contingency coverage.

The total cost baseline is **AUD 500,000**, which is equal to the total approved budget and the total cost breakdown. This ensures consistency between the Project Charter, cost breakdown, and cost baseline.

Funding requirements and sources

The Smart Retail Inventory and Demand Prediction System will be funded through the organisation's internal digital transformation budget. This funding source is appropriate because the project is directly aligned with the organisation's strategic objective of improving operational efficiency, increasing forecast accuracy, reducing out-of-stock incidents, reducing excess inventory holding costs, and supporting long-term digital transformation.

Funding will be released in phases according to project milestones. The first funding release will support project initiation, planning, system design, and early cloud setup. The second release will support AI/ML model development, architecture completion, and initial integration work. The third release will support system integration, cloud operation, testing, and quality assurance. The final release will support UAT, training, go-live, hypercare, and project closure.

Vendor-related expenditure, including cloud hosting, API tools, software services, and external support, will be managed through milestone-based payment arrangements where possible. This ensures that payment is linked to completed deliverables and reduces the risk of paying for incomplete or unacceptable work. The Finance Department will review project spending monthly, while the Project Manager will monitor expenditure weekly against the approved cost baseline.

The contingency reserve of **AUD 50,000** will be used only for approved project risks, such as unexpected integration issues, additional cloud scaling requirements, testing defects, vendor delays, or deployment support needs. Use of contingency funding must be documented and approved according to

the project governance process. Any cost increase beyond the approved **AUD 500,000** budget must be submitted as a formal change request and approved by the Project Sponsor.

Justification, Tools, Time, and Resource Estimation

Strategic justification of the IT project

Why this project?

The smart retail inventory and demand prediction system can help address several challenges that retail organizations face. The company currently relies on a traditional rule-based inventory management system across different ERP, POS, and warehouse management systems. This fragmented infrastructure can lead to a poor inventory, resulting in inaccurate demand forecasting, delayed decisions, and inefficient stock management across various locations.

So, the main proposed IT initiative aims to help to solve these challenges by using an AI-driven inventory forecasting and a demand prediction system. By using machine learning models like LSTM and XGBoost, the system can analyze the sales data based on the history, customer purchasing patterns, and current trends. So, using the real-time inventory information, we can generate the accurate demand forecast. This will allow the organization to make a very proactive inventory decision instead of relying on a manual process.

The system will also aim to integrate with the company's existing ERP and POS platforms through secure APIs to provide real-time inventory visibility and tracking. Automated purchase order generation and dynamic safety stock calculation will help to minimize human error, reduce stock outages, and improve the inventory turnover rates across various retail locations.

Currently, it is a very right time to implement this solution in the retail industry. This is because it is becoming extremely competitive and increasingly data-driven. Customers now expect a much faster product availability, accurate order fulfillment, and a very seamless shopping experience, both in the digital and physical platforms. All the other competitors are also adopting to these AI-powered retail technologies in order to improve operational efficiency and strengthen supply chain performance. Without any digital transformation, the organization has a risk of falling behind and losing market shares.

So, the project strongly aligns with the strategic goals of the company to improve efficiency, reduce inventory holding costs, and enhance customer satisfaction by supporting long-term digital transformation initiatives. By improving these accuracies, the system is expected to reduce the stock fallout incidents, minimize the excessive inventory, and improve decision-making, basically having an overall profitability.

Additionally, the project will also support scalability and future chances for the business's growth by providing a cloud-based infrastructure which is capable of supporting various retail locations with a larger data set and future AI improvements. Through these automations, data analytics, and real-time inventory management, the smart retail inventory and demand prediction systems will also create long-term business value and strengthen the competitive position of the organization in this market sector.

Alignment with Business Strategy

The smart retail inventory and demand prediction system aligns with the organization's objectives, which is to improve operational efficiency, improve customer satisfaction, and also reduce the operational cost. Not only that, the organization aims to support a long-term digital transformation. As the retail industry is becoming increasingly very competitive and also data-driven, every organization must adopt these advanced technological changes so that they remain in the competition and meet with

the evolving requirements of each customer. The proposed system helps to support this transition by enabling predictive decision-making, automated inventory management, and real-time operational visibility across all various retail locations.

One of the most important objectives that is supported by this project is the improvement of operational efficiency. So the current inventory management approach basically relies heavily on the decision-making and the divided systems, which often results in stockouts, overstocking, or a delayed replenishment process. So by implementing the AI-driven forecasting model, the organization can improve the accuracy of this demand prediction system and help to avoid these problems in the future. This will also reduce the manual workload, minimize the operational inefficiencies, and improve the supply chain responsiveness.

The project also helps to support customer satisfaction and thus builds trust towards the company. Accurate inventory forecasting and real-time inventory visibility help to ensure that the project remains available during periods when there is a high customer demand. So reducing the stockout incidents will improve the customer experience and satisfaction, thus building the customer's trust and further strengthening the brand's reputation. In a modern retail environment, product availability and efficient order fulfillment are very critical factors to influence a customer and have a competitive advantage.

The project supports the organization's roadmap to digital transformation by introducing a cloud-based infrastructure and AI technologies, also advanced analytical capabilities. The digital transformation initiative will provide a scalable technology foundation that will support the future business growth, expansion in retail locations, and further AI-driven innovations. The project delivers a very clear value proposition by improving the forecasting accuracy, reducing the operational inefficiencies, and also enabling proactive inventory management. Seeing it from a financial perspective, the organization is expected to achieve a strong return on investment through the reduced stock losses, improved operational productivity, lower inventory holding costs, and increased sales opportunities, resulting in better product availability.

The expected project outcomes include:

- Reduction in stockout incidents
- Improved inventory turnover
- Increased forecast accuracy
- Lower excess inventory costs
- Enhanced operational efficiency
- Improved customer satisfaction
- Better data-driven decision making

Chosen Systems/Tools/Technologies

System/Technology options considered

Tool/System	Description	Pros	Cons
LSTM (Long Short-Term Memory)	It is a deep learning model that can be used for time-series forecasting and demand prediction by analyzing the historical data of	● Predicts demand trends accurately. ● Reduces stockouts and overstocks. ● Learns sales patterns effectively.	● Requires large amounts of historical data ● Needs high computing power

	customer behaviour and sale trends.	• Supports better inventory planning decisions.	• Complex to develop and maintain • Can increase implementation costs
XGBoost	It is a machine learning algorithm which is used for predictive analytics and retail forecasting. It helps to identify relationships between sales trends and inventory movement.	• Fast and efficient for analysing retail datasets • Improves forecasting accuracy • Scalable for large business operations • Works well with structured inventory data	• Requires technical expertise for setup • Parameter tuning can be time-consuming • Results may be difficult for non-technical users to understand
Cloud Hosting Services	It provides a scalable cloud infrastructure and storage required for real-time inventory monitoring and forecasting operations.	• Reduces need for physical hardware • Supports remote access and scalability • Improves system flexibility and availability • Easier to support future business growth	• Ongoing subscription and hosting costs • Dependent on stable internet connection • Third-party provider downtime may affect operations
API Integration Middleware	It is used to connect ERP, POS, and Warehouse Management Systems with the forecasting platform with the help of APIs.	• Enables real-time data sharing between systems • Reduces manual data entry errors • Improves operational efficiency	• Integration with older systems may be difficult • Setup and testing can take time • Requires strong cybersecurity protection

Cybersecurity and access control system	It is security tool used to protect inventory, operational, and customer data through monitoring and authentication controls.	• Protects sensitive business and customer data • Reduces risk of cyberattacks and data breaches • Supports regulatory compliance • Improves system reliability and trust	• Adds additional implementation costs • Requires regular updates and monitoring • Security setup may increase deployment complexity
Analytics and Dashboard Tools	It is used to visualise forecasting reports, inventory levels, and operational performance.	• Easy for managers to monitor inventory in real time • Improves decision-making using visual reports • User-friendly and accessible • Helps track operational performance quickly	• May require staff training • Dashboard customisation can increase development time • Ongoing maintenance and updates may be needed

Final Selection

After evaluating the multiple technologies and options for the systems, the organization has selected a combination of LSTM, which is the long short-term memory, XGBoost, cloud-based infrastructure, and API integration middleware for the smart retail inventory and demand prediction system. These technologies have been chosen because they provide a very strong forecasting accuracy, scalability, and compatibility of the system with long-term operational values for the organization's retail operations.

LSTM has been selected as one of the primary forecasting models because we know it is highly effective for analyzing the time series details of sales data and identifying the long-term demand patterns. Since retail demand is heavily influenced by seasonal trends, customer purchasing behavior, and market fluctuations, LSTM will provide a better forecasting accuracy compared to traditional statistical forecasting methods. So, this can allow the organization to make a very proactive inventory decision and reduce the stock shortage or overstock situations. Now,

XGBoost has been selected to complement the LSTM forecasting model because it has the speed, efficiency, and ability to process the structured retail data as required.

XGBoost can help to improve the predictive performance by identifying the complex relationships between demand fluctuations and sales variables, also inventory levels. Combining these two can provide a very hybrid forecasting approach and can improve the accuracy and operational reliability. A cloud-based infrastructure has been selected because it offers scalability and cost efficiency for the organization. Cloud hosting can also allow the system to support multiple retail locations, large data sets, and real-time inventory processing without requiring major on-premises hardware investments. The environment also supports business expansion and remote system access for operational staff and

management teams. Compared to that of a traditional on-premise infrastructure, it reduces the maintenance cost and improves availability and disaster recovery capabilities.

Now, API integration middleware has been selected as the communication between the smart retail inventory and demand prediction system and the organization's existing ERP, POS, and warehouse management system. Since the organization currently operates on multiple disconnected systems, this is very essential for enabling a real-time data exchange, an automated replenishment process, and accurate inventory visibility across all retail locations. The middleware solution also minimizes manual data entry and reduces operational inefficiencies.

The technologies that have been selected provide a strong cost-benefit advantage for organizations. Although the initial implementation costs are relatively high due to the AI development, integration activities, and cloud infrastructure setup, the long-term operational savings are expected to outweigh these expenses. Improved forecasting accuracy will help to reduce excessive inventory holding costs, minimize the stockout incidents, improve the inventory turnover, and increase customer satisfaction. Thus, it is expected to generate a positive return on investment over time.

The selected systems provide a strong compatibility and usability advantage. Cloud-based dashboards and automated forecasting tools improve usability for store managers, team, and the inventory controller by providing real-time inventory tracking and data-driven insights. The technologies are also very compatible with the existing EIPPOS and WMS systems, which reduces the need for a complete replacement of the infrastructure.

Overall, selected technologies can provide a very scalable, efficient, and future-ready solution to support the organization's digital transformation strategies and long-term operational goals.

Time, Cost and Resource Estimation

Project Duration

Phase	Description	Start Date	End Date	Total Duration	Buffer days
Project Initiation	The main objective is Project Planning, requirement Identification and Stakeholder approval	26 February, 2026	04 Mar, 2026	1 Week	2 Days
Requirement Gathering	All the requirements are gathered which involves operational, functional and business type requirements.	5 March, 2026	25 March, 2026	3 Week	3 Days
System Design & AI Development	This phase involves the designing of predicted architecture and development of the LSTM and XG Boost Model.	26 March, 2026	29 April, 2026	5 Week	5 Days
System Integration	Involves the integration of ERP,	30 April, 2026	13 May, 2026	2 Week	3 Days

	POS, and WMS systems using APIs.				
Testing & Quality Assurance	In this phase different testings are done which involves functional, system and user acceptance testing.	14 May, 2026	03 June, 2026	3 Week	4 Days
Training & Deployment	Involves staff training, onboarding and the deployment of the project.	04 June, 2026	23 June, 2026	3 Week	2 Days

Budget Breakdown

Category	Estimated Cost (AUD)	Notes
System Design & Architecture	60,000	Technical planning, forecasting architecture, integration design, and dashboard design.
AI/ML Model Development	120,000	Development of LSTM and XGBoost models for demand prediction.
System Integration & Development	100,000	Integration of ERP, POS, and Warehouse Management Systems, including API middleware and replenishment logic.
Cloud Infrastructure & Hosting	80,000	Cloud servers, storage, compute resources, hosting, backup, and monitoring.
Testing & Quality Assurance	50,000	Functional testing, integration testing, system testing, UAT, and defect management.
Project Management, Training & Change Management	40,000	Project coordination, stakeholder meetings, governance reporting, staff training, onboarding, and change support.
Contingency Reserve	50,000	Reserved for unexpected technical, operational, integration, or deployment issues.
Total Budget	500,000	Approved project budget.

Resource Planning

Phase	Resource	Allocation	Notes
Project Initiation	Project Manager	Full-Time	Oversees project planning, scheduling, communication, and governance activities.
Requirement Gathering	Business Analyst	Full-Time	Collects operational and technical requirements from stakeholders and retail teams.
System Design	Technical Lead and Developer	Full-Time	Designs forecasting architecture, dashboards, APIs, and database structure.
AI Model Development	ML Engineers	Full-Time	Develops and trains LSTM and XGBoost forecasting models for demand prediction.
System Integration	Software Developers	Full-Time	Integrates ERP, POS, and Warehouse Management Systems (WMS) using APIs.
Cloud Infrastructure Setup	Cloud Engineer	Part-Time	Configures cloud hosting, storage systems, backup, and network infrastructure
Testing and Quality Assurance	QA Test Team	Part-Time	Conducts functional testing, integration testing, and user acceptance testing.
Cybersecurity Implementation	Cybersecurity Specialist	Part-Time	Implements security monitoring, access control, and data protection measures.
Training and Deployment	Change Management Team	Full-Time	Provides staff training, onboarding support, and deployment assistance.
Post- Go-Live Support	Help Desk Team	Part-Time	Resolves technical issues and provides operational support after system deployment

Risk Management Plan

Introduction

Risk management is an essential part of planning and delivering the Smart Retail Inventory and Demand Prediction System. This project carries uncertainty in several areas, especially machine learning performance, data quality, cloud reliability, cybersecurity, vendor coordination, and integration with the existing ERP, POS, and WMS systems. If these issues are not addressed early, they may cause delays, increase project costs, reduce system quality, or weaken user acceptance. For this reason, risk management should be treated as an ongoing project activity, not only as a planning task completed at the beginning.

The purpose of this Risk Management Plan is to set out a practical process for managing project risks throughout the project lifecycle. It explains how the project team will identify potential problems, assess their likely impact, and decide on suitable response actions. It also clarifies how responsibility for each risk will be assigned and how the effectiveness of mitigation actions will be reviewed over time. This is particularly important because the project depends on both technical delivery and organisational adoption. For example, poor data quality may reduce forecast accuracy, delays from vendors may affect integration work, and limited user acceptance may reduce the value of the system after deployment.

The plan also recognises that uncertainty can sometimes create opportunities. Improved monitoring may strengthen system reliability, while automated purchase order generation may reduce manual procurement effort more quickly than expected. Where these opportunities support the project objectives, they should be recorded and managed in the same way as other project risks. Overall, this Risk Management Plan helps the project team control uncertainty, respond to issues in a timely way, and improve the likelihood of successful deployment and adoption.

Risk Management Approach

The project will use an ongoing risk management approach based on PMBOK risk management principles. Risk management will not be treated as a task completed only during planning. It will be built into the whole project lifecycle, beginning at initiation and continuing through planning, development, testing, deployment, and post-go-live hypercare.

Risk identification will begin during project planning and continue as the project progresses. The first set of risks will be developed by reviewing key project documents, including the Project Charter, Scope Management Plan, Schedule Management Plan, Stakeholder Management Plan, and Project Governance and Roles document. The project team will also identify risks through stakeholder workshops, interviews, previous lessons from similar IT projects, and consultation with both technical and business teams. In this project, particular attention will be given to risks linked to AI/ML model performance, data quality, cloud cost uncertainty, legacy system integration, vendor delays, cybersecurity compliance, and user resistance.

Risk analysis will mainly use qualitative assessment, supported by limited quantitative analysis where the impact can be reasonably estimated. Each risk will be assessed using a probability-impact matrix to determine its likely effect on project objectives. Based on this assessment, risks will be rated as Low, Medium, High, or Extreme. Quantitative analysis will only be used for major cost or schedule risks, such as cloud resource overrun, vendor delay, or API integration failure, where the likely time or financial impact can be estimated with enough confidence.

Risk response planning will define suitable actions for major risks. Negative risks may be avoided, reduced, transferred, accepted, or escalated depending on their severity and the level of control available to the project team. Positive risks will also be considered where they could improve project outcomes, such as faster procurement automation or better system reliability. Each significant risk will be assigned to a risk owner, who will monitor early warning signs, coordinate response actions, and provide updates to the Project Manager. The risk register will be reviewed weekly by the project team and monthly by the Steering Committee to ensure that risk information remains current and useful.

Risk Identification

Risk identification will be conducted as an iterative process throughout the project lifecycle. The first risk identification cycle will occur during project planning, using the approved project scope, schedule, stakeholder analysis, and governance structure as inputs. Since the system depends on multiple technical and organisational components, the project team will use a Risk Breakdown Structure to ensure that risks are considered systematically across technology, data, people, financial, vendor, compliance, and process categories.

Several techniques will be used to capture risks. Brainstorming workshops will be held with the Project Manager, Technical Lead, Business Analyst, QA and Change Management Lead, IT Department, and Inventory and Procurement representatives. These workshops will be useful for identifying broad delivery risks, including integration delays, user resistance, and unclear requirements. Structured interviews will be conducted with subject matter experts, including data analysts, store operations representatives, finance staff, and vendor contacts, to uncover risks that may not be visible to the core project team. SWOT analysis will also be applied to identify both threats and opportunities, such as the risk of poor model accuracy and the opportunity to improve operational decision-making through predictive analytics.

The risk register will be treated as a living document. New risks may be added during sprint reviews, status meetings, quality reviews, UAT sessions, procurement reviews, or change control discussions. Any risk with potential impact on scope, schedule, budget, system quality, compliance, or adoption will be recorded, assigned an owner, analysed, and monitored.

Risk Analysis

Risk analysis will be used to prioritise risks and determine the level of management attention required. The project will primarily use qualitative risk analysis because many risks in AI-enabled IT projects, such as user resistance, model uncertainty, and integration complexity, cannot be estimated precisely at the beginning of the project. Each identified risk will be assessed according to two dimensions: likelihood of occurrence and impact on project objectives.

Likelihood will be rated as Low, Medium, or High. Impact will also be rated as Low, Medium, or High, based on the potential effect on cost, schedule, scope, quality, security, compliance, or stakeholder acceptance. The combination of likelihood and impact will determine the overall severity level. Risks with both high likelihood and high impact will be classified as Extreme and escalated to the Steering Committee. High-severity risks will require active mitigation and weekly monitoring. Medium risks will require assigned ownership and periodic review, while Low risks will be recorded and monitored through normal project reporting.

Where possible, quantitative analysis will be applied to risks with measurable cost or schedule consequences. For example, cloud GPU cost overrun may be estimated using expected usage variance, while vendor delay may be assessed by estimating the number of affected work packages on the critical path. This limited quantitative analysis will support decisions about contingency reserves, schedule buffers, procurement alternatives, and escalation thresholds. The output of risk analysis will be an updated risk register that ranks risks by priority and provides input into response planning.

Risk Response Planning

Risk response planning defines how the project team will address prioritized risks before they threaten project success. For negative risks, the project will use five response strategies: avoid, mitigate, transfer, accept, and escalate. Avoidance will be used where the project can remove a risk by changing scope or approach, such as excluding non-essential dashboard features from the first release. Mitigation will be the most common strategy and will be applied to reduce either the likelihood or impact of risks, such as conducting early data profiling to reduce model training delays. Transfer will be applied where risk can be shifted contractually to vendors through service-level agreements, warranties, or support obligations. Acceptance will be used for low-priority risks where response cost exceeds the likely impact. Escalation will be used where a risk exceeds the Project Manager's authority, such as major budget increase, regulatory change, or enterprise-wide cybersecurity exposure.

For positive risks, the project will use exploitation, enhancement, sharing, acceptance, and escalation. For example, if early model testing achieves higher-than-expected forecast accuracy, the team may exploit the opportunity by accelerating dashboard demonstration activities and stakeholder engagement. If AI-powered monitoring tools reduce post-deployment support risk, the project may enhance this opportunity by integrating monitoring earlier in the deployment plan.

Each major risk will have a risk owner responsible for monitoring triggers and executing response actions. The Project Manager will maintain the overall risk register, review risk status in weekly project meetings, and escalate High or Extreme risks to the Steering Committee. Response actions will be reviewed for cost-effectiveness, feasibility, and alignment with the project's scope, schedule, and quality baselines.

Risk Register

Risk No.	Risk Name	Risk Description	Risk Owner	Mitigation Plan	Impact Level	Likelihood of Occurrence	Severity Level	Contingency Plan
R1	Poor Data Quality	Historical POS, WMS, and ERP data may contain missing values, duplication, inconsistent SKU	Data Analytics Team / Technical Lead	Conduct early data profiling, cleansing, validation rules, and sample	High	High	Extreme	Extend data preparation window, use fallback rule-based forecasting for affected



		codes, or inaccurate inventory records, reducing ML forecast accuracy.		reconciliation before model training.				SKUs, and escalate data ownership issues to the Project Sponsor.
R2	ML Model Accuracy Below Target	LSTM/XGBoost models may fail to achieve the target forecast accuracy of $\geq 85\%$ due to insufficient data quality, seasonality, sparse SKU history, or cold-start items.	Technical Lead / ML Engineer	Use baseline models, feature engineering, cross-validation, error analysis, and cold-start heuristics.	High	Medium	High	Reduce initial model scope to high-volume SKUs, deploy hybrid rule-based + ML forecasting, and continue model improvement during hypercare.
R3	Legacy ERP/POS/WMS Integration Delay	API development and integration with legacy systems may take longer than expected due to undocumented interfaces, vendor constraints, or technical	Technical Lead / IT Department	Conduct early interface discovery, confirm API documentation, create integration prototypes, and hold regular vendor	High	High	Extreme	Use temporary batch-file integration, re-sequence non-dependent tasks, and escalate vendor issues through the Steering

		incompatibility.		technical reviews.				Committee.
R4	Vendor Delivery Delay	Cloud provider, ERP vendor, ML solution vendor, or integration consultants may delay required deliverables or technical support.	Project Manager / Vendor Manager	Define clear SLAs, milestones-based deliverables, vendor reporting cadence, and penalty clauses.	High	Medium	High	Activate backup vendor support, reallocate internal resources, or escalate contract non-performance to procurement governance.
R5	Cloud Cost Overrun	GPU training, storage, data transfer, monitoring tools, and cloud-based MLOps services may exceed budget due to consumption-based pricing.	Project Manager / Finance Department	Set cloud budgets, alerts, usage dashboards, reserved capacity where suitable, and approval thresholds for high-cost workloads.	High	Medium	High	Use contingency reserve, reduce training frequency, optimise compute usage, or move non-critical workloads to lower-cost resources.

Quality Management Plan

Introduction

Quality in this project refers to the degree to which the Smart Retail Inventory and Demand Prediction System meets documented requirements, stakeholder expectations, and its intended business purpose. In IT projects, quality is not limited to whether the software functions correctly; it also includes reliability, performance, usability, maintainability, and security. This aligns with the Week 05 lecture definition that quality concerns whether a product or service meets requirements, expectations, and is fit for its intended purpose.

For this project, quality management is essential because the system will directly affect inventory forecasting, replenishment decisions, supplier coordination, and daily store operations. Poor quality could result in inaccurate demand forecasts, data synchronisation errors, operational disruption, or low user trust. Therefore, this Quality Management Plan covers both product quality and process quality. Product quality focuses on whether the system deliverables meet functional, performance, security, and usability standards, while process quality focuses on whether development, testing, reporting, and approval activities are performed consistently and effectively. The plan includes both quality assurance, which is process-oriented and preventive, and quality control, which is product-oriented and verification-based.

Objective

The objective of this Quality Management Plan is to define how quality will be planned, managed, measured, and controlled throughout the Smart Retail Inventory and Demand Prediction System project. First, it establishes clear quality standards and measurable criteria for major deliverables, including the AI forecasting engine, API integration middleware, operational dashboard, automated replenishment engine, and project documentation. Second, it ensures that quality is built into the project from the beginning through preventive activities such as process audits, peer reviews, coding standards, requirements traceability, and risk-based defect prevention. Third, the plan defines quality control activities that verify whether completed deliverables meet the agreed requirements before acceptance. These activities include unit testing, integration testing, system testing, performance testing, security checks, and User Acceptance Testing. This follows the Week 05 lecture distinction between managing quality as a preventive process and controlling quality as a verification process. Finally, the plan aims to support stakeholder satisfaction by ensuring that the delivered system is reliable, secure, usable, and aligned with the organisation's business objectives.

Scope

The scope of quality management covers both the outputs produced by the project and the processes used to produce them. In terms of product deliverables, quality will be measured across the AI demand forecasting module, data ingestion and ETL pipeline, API integration middleware, automated replenishment engine, web-based operational dashboard, MLOps retraining pipeline, security controls, and user-facing documentation. The AI forecasting module will be assessed against forecast accuracy and model validation criteria. The integration middleware will be assessed against data integrity, latency, availability, and successful synchronisation with ERP, POS, and WMS systems. The dashboard and

replenishment engine will be assessed against functional correctness, usability, exception handling, and business rule accuracy.

The quality scope also includes project management deliverables such as the Project Management Plan, Requirements Traceability Matrix, test plans, defect logs, UAT sign-off records, training materials, and post-implementation review documentation. These documents are important because they provide evidence that project work has been planned, tested, reviewed, and formally accepted.

In terms of process quality, the plan covers development practices, testing procedures, change control, defect management, reporting, and stakeholder review activities. The Week 05 lecture emphasises that project quality management includes both the quality of deliverables and the quality of project processes. Therefore, this plan will not only check whether the final system works, but also whether the team follows agreed standards, conducts appropriate reviews, maintains traceability, and resolves defects before go-live.

Quality Standards and Criteria

The quality standards for this project are based mainly on the project quality management principles covered in the Week 05 lecture. The first applicable standard is ISO 9001, which provides a general quality management system approach focused on continuous improvement, customer satisfaction, and documented processes. In this project, ISO 9001 principles will be reflected through formal requirements documentation, quality metrics, review checkpoints, and corrective action processes.

The second relevant standard is ISO/IEC 27001, because security is a core dimension of IT quality. Since the system processes retail operational data and integrates with ERP, POS, and WMS platforms, quality must include confidentiality, integrity, and availability of data. Security-related criteria will include role-based access control, encryption in transit and at rest, secure API communication, and no critical data leakage or corruption.

CMMI is also relevant because it focuses on assessing and improving software development processes. For this project, the team should aim for a managed and defined process, where requirements, testing, defect resolution, and deployment activities are documented and repeatable. ITIL principles are also applicable after deployment because the system will become an operational IT service. ITIL supports service quality through service level agreements, continuous improvement, and alignment with business needs.

The measurable quality criteria are as follows: the forecasting module should achieve at least 85% SKU-level forecast accuracy; API data synchronisation should have zero critical data loss or corruption; system uptime should reach at least 99.5% after deployment; API response latency should remain below the agreed threshold during peak operating hours; all Severity 1 and Severity 2 defects must be resolved before go-live; and UAT sign-off must be obtained from key business users. These criteria align with the project success criteria stated in the Project Charter.

Quality Assurance and Control Activities

Quality assurance activities will focus on preventing defects and ensuring that project processes are followed correctly. The QA Lead will maintain the test strategy, define quality metrics, conduct process audits, and ensure that development and testing practices comply with the approved Quality Management

Plan. Process audits will check whether developers follow coding standards, whether user stories include acceptance criteria, whether test cases are linked to requirements, and whether CI/CD logs and defect records are maintained properly. This reflects the Week 05 lecture explanation that managing quality translates the Quality Management Plan into executable quality activities and focuses on preventing defects rather than only detecting them.

Quality assurance will also include peer code reviews, architecture review sessions, sprint retrospectives, requirements traceability checks, and security design reviews. These activities are intended to identify process weaknesses early, reduce technical debt, and improve collaboration between the development, QA, business, and operations teams. For example, the Requirements Traceability Matrix will be used to ensure that each approved requirement is linked to a design component, test case, and acceptance result.

Quality control activities will focus on inspecting and validating completed deliverables. The testing team will perform unit testing for individual software modules, integration testing for interfaces between the AI engine and ERP/POS/WMS systems, system testing for the complete solution, performance testing for API latency and system responsiveness, security testing for access control and data protection, and User Acceptance Testing with Store Managers and the Procurement Team. The Week 05 lecture identifies unit testing, integration testing, system testing, and UAT as key testing types for quality control. Defects will be logged, prioritised by severity, assigned to owners, retested after correction, and reviewed before acceptance. Deliverables will only be accepted when they satisfy the defined quality criteria and receive formal sign-off.

Communication Management Plan

Objectives

Communication is necessary for ensuring that project information is delivered to the right stakeholders, through the right channels, at the right time and in the right format. In the Smart Retail Inventory and Demand Prediction System, effective communication is particularly important because the project involves multiple internal and external stakeholders, including executive sponsors, IT teams, inventory and procurement users, store managers, cloud providers, ML vendors, ERP vendors and system integration consultants. These stakeholders have different levels of power, interest and information needs, so a single informal communication approach would not be sufficient.

The purpose of this Communication Management Plan is to define how project information will be collected, distributed, stored, retrieved and monitored throughout the project lifecycle. This aligns with the project communication management process covered in the lecture slides, where communication planning identifies stakeholder communication needs, communication methods, roles, responsibilities, frequency and feedback mechanisms. The plan also supports managing and monitoring communications by ensuring that information flow remains accurate, timely and aligned with stakeholder expectations.

This plan will help reduce misunderstanding, duplicated work, delayed decisions and stakeholder resistance. It will also support project governance by creating clear reporting lines, escalation paths and documentation standards for project status, risks, issues, changes, testing, deployment and post-go-live support.

Scope

The scope of this Communication Management Plan covers all formal project communications required to support the planning, execution, monitoring, control and closure of the Smart Retail Inventory and Demand Prediction System. It applies to communications between the project team, executive sponsor, steering committee, business users, technical teams, vendors and other relevant stakeholders. The plan includes communication related to project status reporting, schedule updates, budget updates, risk and issue escalation, change requests, quality and testing updates, UAT coordination, training notices, vendor coordination, deployment readiness and post-implementation review.

This plan does not cover informal personal conversations, vendor-internal communications, unrelated organisational announcements, or detailed technical documentation that is managed under separate technical workstreams. For example, source code documentation, system architecture specifications, detailed API documentation and user manuals may be communicated through this plan, but their technical content is managed separately by the relevant technical owners.

The Communication Management Plan supports the overall Project Management Plan by linking stakeholder engagement, governance, schedule control, risk management, quality management and change management. In the project lifecycle, planning defines the

roadmap, execution implements the work and communication plan, and monitoring and controlling tracks performance, risks, quality, resources and communications. Therefore, this plan ensures that communication is not treated as an ad-hoc activity, but as a controlled management process across the entire project lifecycle.

Stakeholder information requirements

The information requirements for each stakeholder are based on their level of interest, project role and degree of influence. The Project Sponsor and Steering Committee require high-level information about project progress, budget status, milestone achievement, major risks, unresolved issues and decisions requiring executive approval. Their communication should focus on governance, strategic alignment and exception-based reporting rather than operational detail.

The Project Manager requires detailed and continuous information from all workstreams, including scope progress, schedule performance, cost status, resource issues, vendor updates, risks, defects and stakeholder concerns. This information is needed to coordinate delivery, maintain the project baseline and escalate issues when required. The Technical Lead, ML Engineer, IT Department and Data Analytics Team require technical information such as infrastructure readiness, data quality, model performance, integration dependencies, testing defects and deployment constraints. Their communication must be frequent because technical delays or integration issues can directly affect project schedule and quality.

The Inventory and Procurement Team and Store Managers are primary business users. They require information about requirements validation, workflow changes, UAT activities, training arrangements, go-live impacts and support channels. Their communication should be clear, practical and focused on adoption because poor user understanding may lead to resistance or inconsistent system use. Finance requires budget and cost variance information, while external vendors require technical specifications, delivery expectations, issue logs, service levels and formal change notices. Regulatory or compliance stakeholders require milestone-based evidence that privacy, security and compliance obligations are being addressed.

Communication Table

Stakeholder Name	Level of Interest	Description of interest	Document Name	Communication Medium	Frequency of communication
Project Sponsor	High	Requires visibility of project progress, budget status, major risks, milestone performance and strategic alignment.	Executive Status Report; Steering Committee Brief	Steering Committee meeting; email report	Monthly; immediate escalation for critical issues
Steering Committee	High	Requires governance-level updates and decisions on major scope, cost, schedule, risk or vendor issues.	Steering Committee Report; Decision Log	Formal meeting; presentation; email summary	Monthly

Project Manager	High	Requires detailed information across all workstreams to coordinate delivery, manage baselines and control risks/issues.	Project Status Report; Risk Register; Issue Log; Change Log	Jira/Confluence; Teams; email; meetings	Daily/weekly depending on project phase
Business Analyst	High	Requires stakeholder feedback, requirements clarification, process change information and UAT inputs.	Requirements Register; RTM; UAT Feedback Log	Workshops; Teams; email	Weekly; more frequently during requirements and UAT
Technical Lead / ML Engineer	High	Requires technical requirements, model performance updates, integration dependencies, architecture decisions and defect information.	Technical Progress Report; Model Validation Report; Integration Log	Daily stand-up; Jira; technical meetings	Daily during development ; weekly otherwise
IT Department	High	Requires infrastructure, security, access control, cloud environment, deployment and operational support information.	Infrastructure Plan; Security Checklist; Deployment Plan	Technical workshops; Teams; email	Weekly; ad-hoc for incidents or deployment issues
Inventory & Procurement Team	High	Requires requirements validation, replenishment workflow updates, UAT instructions and training information.	Requirements Document; UAT Plan; Training Materials	Workshops; UAT sessions; email	Fortnightly; weekly during UAT and training

Store Managers	High	Require information about operational impacts, go-live timing, training arrangements, support channels and dashboard use.	Go-Live Notice; Training Schedule; Quick Reference Guide	Email; training workshops; project newsletter	Fortnightly; weekly before go-live
Data Analytics Team	High	Requires data quality updates, forecasting model validation results and reporting/dashboard requirements.	Data Quality Report; Model Validation Report	Technical meeting; shared repository; Teams	Weekly during data and model phases
Finance Department	Medium	Requires budget usage, cost variance, procurement spending and approval-related financial information.	Budget Status Report; Cost Variance Report	Email report; finance review meeting	Monthly
QA & Change Management Lead	High	Requires defect status, test readiness, UAT progress, user training progress and adoption risks.	Test Report; Defect Log; Change Readiness Report	Jira; Teams; QA meetings	Weekly; daily during SIT/UAT
Cloud Services Provider	Medium	Requires cloud configuration requirements, SLA expectations, support requests and incident escalation information.	Cloud Configuration Document; SLA Report; Support Ticket Log	Vendor meeting; ticketing system; email	Fortnightly; ad-hoc for incidents
ML Solution Vendor	High	Requires model acceptance criteria, data requirements, technical constraints, performance feedback and	ML Requirements Specification; Model Test Report; Issue Log	Vendor meeting; Jira; email	Weekly during ML development

		issue resolution information.			
ERP Software Vendor	Medium	Requires API specifications, integration schedules, testing windows, defect reports and change notices.	API Integration Specification; Integration Test Report	Technical meeting; vendor portal; email	Fortnightly; weekly during integration testing
System Integration Consultants	High	Require architecture decisions, integration dependencies, defect status, deployment readiness and cutover information.	Integration Plan; Defect Log; Deployment Checklist	Workshops; Teams; Jira	Weekly
Regulatory / Compliance Representatives	Low to Medium	Require evidence that privacy, security and compliance obligations are being addressed.	Compliance Report; Security Assessment	Formal documentation ; email	At compliance milestones

Change Management Plan

Objectives

In this project, “change” refers to the organisational, operational, behavioural, and technological transition caused by the implementation of the Smart Retail Inventory and Demand Prediction System. The project does not only introduce a new technical platform; it changes how inventory decisions are made, how procurement staff interact with demand forecasts, how Store Managers monitor stock risks, and how legacy ERP, POS, and WMS data are used in daily operations. Therefore, change management is required to ensure that the organisation can move from reactive, rule-based inventory management to predictive, data-driven inventory operations.

The purpose of this Change Management Plan is to provide a structured approach for preparing stakeholders, supporting adoption, reducing resistance, and sustaining the benefits of the new system. This aligns with the principle that successful IT change depends not only on delivering the technical solution, but also on ensuring that users understand, accept, and consistently use the new way of working. For this project, effective change management is particularly important because the system introduces machine learning-based forecasting, automated replenishment recommendations, and a Human-in-the-Loop procurement console. These features may alter existing responsibilities and require users to trust algorithmic recommendations while still retaining appropriate human oversight. The plan therefore focuses on communication, training, readiness assessment, support channels, resistance management, and post-go-live adoption monitoring.

Scope

The scope of this Change Management Plan covers all change activities required to support the successful adoption of the Smart Retail Inventory and Demand Prediction System across the target organisation. It applies to the 45 Store Managers, the central Inventory and Procurement Team, IT support staff, the Data Analytics Team, and relevant managers who will use, support, or make decisions based on the new system. It also covers indirect stakeholder groups such as Finance, external vendors, and the Project Sponsor where their involvement is required for decision-making, communication, or operational readiness.

The plan includes stakeholder readiness assessment, change communication, user training, support arrangements, adoption monitoring, and resistance mitigation before and after go-live. It also includes the preparation of change enablement materials such as Quick Reference Guides, training modules, workshop materials, FAQs, and post-deployment support procedures. This is consistent with the scope plan, which identifies a Change Management and User Training Asset Library as a formal deliverable designed to support the target adoption rate of at least 90% among operational staff.

However, this plan does not replace formal technical change control for scope, cost, schedule, or architecture changes. Technical changes affecting the approved project baseline will continue to follow the project governance process and the integrated change control procedure. For example, changes exceeding AUD \$20,000, altering project scope, or shifting the go-live date by more than two weeks require formal Sponsor approval through the Tier 3

governance process. Therefore, this plan focuses on people-side adoption and operational transition, while technical baseline changes remain governed by the Change Control Log, Project Manager, Technical Lead, and Steering Committee.

Impact Assessment

The current state of IT in the organisation is characterised by fragmented, reactive, and rule-based inventory management. The organisation currently operates across 45 retail locations and relies on siloed ERP, POS, and WMS systems, which limits real-time visibility and contributes to stockouts, overstocking, and inefficient procurement decisions. Inventory decisions are largely based on historical reports, manual judgement, and static replenishment thresholds rather than predictive analytics.

The future state will be a cloud-native, AI-enabled inventory and demand prediction environment. The system will integrate POS, WMS, and ERP data through secure APIs, generate SKU-level demand forecasts using LSTM and XGBoost models, support automated purchase order generation, and provide a centralised operational dashboard with predictive alerts and Human-in-the-Loop approval functions. This creates several major impact areas.

First, business processes will change from manual and retrospective inventory review to proactive, forecast-driven replenishment. Procurement staff will spend less time calculating routine order quantities and more time reviewing exceptions, validating forecasts, and managing supplier constraints.

Second, roles and responsibilities will change. Store Managers will be expected to monitor dashboard alerts, validate store-level stock risks, and provide feedback on local demand conditions, while the Inventory and Procurement Team will become primary users of the HITL console. The QA and Change Management Lead will also coordinate training and adoption activities, as already defined in the governance plan.

Third, technology adoption will require users to understand AI-generated forecasts, dashboard indicators, system alerts, and approval workflows. The main challenge is not simply technical access but user confidence in algorithmic recommendations.

Finally, organisational culture will shift toward data-driven decision-making. This may create tension for staff who are used to experience-based ordering or who fear that automation will reduce their decision authority.

Training and Support Plan

Training will be role-based and delivered before go-live to ensure that all affected users can perform their required tasks in the new system. Store Managers will receive practical training on dashboard navigation, inventory alerts, stockout and overstock risk interpretation, and escalation procedures. The Inventory and Procurement Team will receive deeper training on forecast review, HITL approval workflows, automated purchase order generation, exception handling, and override documentation. IT support staff will receive training on access management, incident logging, troubleshooting, and integration monitoring.

Training will be delivered through half-day workshops, guided demonstrations, scenario-based exercises, Quick Reference Guides, and online learning materials. This is consistent with the project schedule, which includes half-day training workshops and Quick Reference Guides before go-live. During the post-deployment Hypercare period, support will be provided to resolve production issues, answer user questions, and reinforce correct usage. Support channels will include a Service Desk ticketing process, a dedicated Microsoft Teams support channel, FAQs, knowledge base articles, and escalation to the Technical Lead or QA and Change Management Lead for complex issues.

Resistance Management Strategy

The main sources of resistance are expected to come from Store Managers and procurement staff. Store Managers may worry that the system will disrupt existing store-level routines or reduce local decision-making flexibility. Procurement staff may be concerned that automated replenishment will replace manual judgement or reduce the importance of their role. Some users may also distrust AI-generated forecasts, particularly when recommendations conflict with their experience or when demand patterns are affected by local events.

To mitigate these risks, the project will use early engagement, transparent communication, and practical involvement in UAT. Users will be shown that the system supports decision-making rather than fully replacing human judgement. The Human-in-the-Loop console will be emphasised as a control mechanism that allows authorised staff to review, adjust, and approve automated recommendations. Change champions will be selected from the Inventory and Procurement Team and Store Manager group to provide peer support and collect feedback. Resistance will also be monitored through training attendance, UAT feedback, support tickets, adoption dashboards, and post-go-live surveys. Where resistance is linked to misunderstanding, additional training will be provided; where it is linked to workflow concerns, the issue will be escalated through the change and governance process.

Monitoring and Review

Adoption and change success will be monitored using both quantitative and qualitative KPIs. The primary adoption KPI is the target system usage rate of at least 90% across all intended users within 60 days of launch, which aligns with the project's business value and adoption objectives. Additional KPIs include training completion rate, user satisfaction score, number of unresolved support tickets, average incident response time, number of manual override actions, dashboard login frequency, and percentage of procurement decisions processed through the new system.

Operational KPIs will also be reviewed to determine whether the change is delivering business value. These include forecast accuracy, stockout reduction, excess inventory holding cost reduction, API reliability, and system uptime. Review activities will occur weekly during the post-deployment Hypercare period and monthly during the first three months after deployment. The Project Manager, QA and Change Management Lead, IT Department, and Inventory and Procurement Team will review adoption data, support trends, and user feedback. Lessons learned will be documented and used to update training materials, support procedures, and future change initiatives.

Project Monitoring and Control Plan

Introduction

Project monitoring and control is critical for the Smart Retail Inventory and Demand Prediction System because the project involves multiple interdependent workstreams, including cloud infrastructure, historical data migration, AI/ML forecasting model development, API integration with ERP/POS/WMS systems, testing, user training, and go-live support. In IT project management, monitoring and controlling ensures that the project remains aligned with its approved scope, schedule, cost, quality standards, and stakeholder expectations throughout execution. This is particularly important for this project because delivery delays or uncontrolled changes may affect the planned go-live date, integration readiness, user adoption, and business benefits such as improved forecast accuracy and reduced stockouts.

This plan establishes a structured approach for tracking project performance, identifying variances against approved baselines, escalating issues, and implementing corrective or preventive actions. It also ensures that any proposed change to scope, time, cost, quality, or resources is assessed through formal change control rather than being introduced informally during development.

Objective

The main objective of this Project Monitoring and Control Plan is to provide a consistent mechanism for measuring project performance and ensuring that the project remains on track against its approved baselines. Specifically, the plan aims to monitor progress against the Work Breakdown Structure, project schedule, approved budget, quality targets, risk register, issue log, and stakeholder engagement requirements. This supports early identification of deviations before they become major delivery problems.

The plan also aims to improve decision-making by ensuring that project performance data is collected, analysed, and reported regularly to the Project Manager, project team, and Steering Committee. Since this project has defined success criteria, including $\geq 85\%$ SKU-level forecasting accuracy, zero critical data loss, $\geq 99.5\%$ uptime, and $\geq 90\%$ user adoption, monitoring activities must connect technical progress with measurable business outcomes. Finally, the plan ensures that corrective actions, preventive actions, and formal change requests are handled in an integrated way across scope, schedule, cost, quality, risk, and stakeholder management.

Scope

The monitoring and control scope covers all major dimensions of project performance, with particular attention to the triple constraints of scope, time, and cost. Scope monitoring will ensure that all approved deliverables, including the AI demand forecasting module, cloud data ingestion pipeline, API middleware, operational dashboard, automated replenishment engine, testing documentation, UAT sign-off, and training materials, are developed according to the approved scope baseline. Any additional feature request, change to acceptance criteria, or modification to system functionality will be assessed through the formal change control process to prevent scope creep.

Schedule monitoring will track progress against the approved milestone plan, including cloud and data infrastructure establishment, AI model and API middleware completion, SIT sign-off, UAT sign-off, go-live, staff training, and project closure. Cost monitoring will compare actual expenditure against the AUD 500,000 approved budget, including allocated funding for system design, AI/ML model development,

integration, cloud infrastructure, testing, project management, training, change management, and contingency reserve. Quality monitoring will focus on test outcomes, defect trends, forecast accuracy, API latency, data integrity, uptime readiness, and UAT acceptance. The plan also monitors risks, issues, resource workload, stakeholder communication, vendor performance, and user adoption readiness.

Monitoring Methods and Tools

The project will use a combination of schedule, cost, scope, quality, risk, and communication monitoring tools to provide a complete view of project performance. For schedule monitoring, a Gantt chart will be used to compare planned activities against actual progress. This is appropriate because the schedule management lecture identifies Gantt charts as a key tool for visualising project timelines, tracking progress against the baseline schedule, and identifying overlaps, leads, and lags. The Project Manager will update the schedule weekly based on completed tasks, delayed tasks, milestone readiness, and dependencies between workstreams such as data engineering, ML model development, API integration, SIT, UAT, and training.

For cost monitoring, the project will apply lightweight Earned Value Management (EVM). Planned Value (PV), Earned Value (EV), and Actual Cost (AC) will be used to calculate Cost Variance (CV), Schedule Variance (SV), Cost Performance Index (CPI), and Schedule Performance Index (SPI). These indicators allow the project team to identify whether the project is under or over budget and whether completed work is consistent with the planned schedule. Because this is a hybrid IT project with iterative AI/ML development, EVM will be combined with sprint-level progress reporting, burn-up/burn-down tracking, and deliverable-based acceptance.

For scope monitoring, the Requirements Traceability Matrix (RTM), WBS, issue log, and change control log will be used to ensure that every approved requirement is mapped to a deliverable, test case, and acceptance result. This helps prevent scope creep and ensures that changes to requirements are formally reviewed. For quality monitoring, the QA & Change

Management Lead will track defect density, severity 1/2 defects, SIT pass rate, UAT completion, API latency, data migration accuracy, forecast model performance, and training completion rate. Risk and issue monitoring will be conducted through weekly risk register reviews, issue log updates, and escalation reports. Communication monitoring will be supported through weekly team stand-ups, fortnightly project status reports, Jira/Confluence dashboards, and monthly Steering Committee reviews.

Control Mechanisms

Control mechanisms will be applied whenever monitoring activities identify a significant deviation from the approved project baseline. The Project Manager will first determine whether the deviation relates to scope, schedule, cost, quality, resources, risk, stakeholder engagement, or procurement. Minor deviations that can be resolved within the approved project tolerances will be managed directly by the relevant workstream owner and reported in the weekly status update. More significant deviations, especially those affecting approved milestones, budget limits, critical deliverables, security compliance, UAT sign-off, or go-live readiness, will be escalated to the Steering Committee.

The control process will follow a structured sequence: identify the variance, analyse the root cause, assess the impact, define corrective or preventive actions, assign an owner, agree on a due date, implement the response, and verify whether the issue has been resolved.

Corrective Actions

Corrective actions will be used when actual project performance has already deviated from the approved baseline and the project needs to be brought back on track. For example, if API integration falls behind schedule, the Project Manager may reassign developers, prioritise critical integration endpoints, or approve limited schedule compression through fast-tracking. If cloud infrastructure spending exceeds the planned budget, the Technical Lead will review instance usage, remove unused resources, and optimise GPU or storage consumption. If SIT identifies severity 1 or severity 2 defects, the QA Lead will create a defect recovery plan, assign fix owners, and require retesting before UAT begins. Corrective actions must be recorded in the issue log and reviewed until closure.

Preventive Actions

Preventive actions will be used to reduce the probability of future deviations before they occur. For this project, preventive actions include weekly schedule variance reviews, early risk register updates, peer review of technical designs, regular data quality checks, cloud cost alerts, test readiness reviews, and stakeholder feedback sessions before UAT. Because AI/ML work can be uncertain due to data quality, model tuning, and forecasting accuracy risks, the project team will use early model validation checkpoints rather than waiting until final testing to confirm performance. Preventive actions will also include maintaining contingency reserve, monitoring resource workload, confirming vendor availability, and ensuring training materials are prepared before go-live to reduce adoption risks.

Change Control Integration

Project monitoring and control will be directly integrated with the Change Management Plan. Any variance that cannot be corrected within the approved baseline will trigger a formal

change request. This includes changes to project scope, delivery schedule, budget allocation, acceptance criteria, technical architecture, data integration requirements, security controls, training approach, or go-live timing. The change request must document the reason for the change, affected deliverables, impact on scope, schedule, cost, quality, risks, stakeholders, and recommended approval pathway.

Minor operational adjustments may be approved by the Project Manager if they do not affect baselines. However, changes with strategic, financial, contractual, or go-live implications must be escalated to the Steering Committee or Project Sponsor. Once approved, the change will be recorded in the Change Control Log, and relevant project documents, including the

schedule, budget, risk register, issue log, RTM, and communication plan, will be updated. This ensures that monitoring results are not treated as isolated observations, but are converted into controlled decisions that preserve project governance and delivery accountability.

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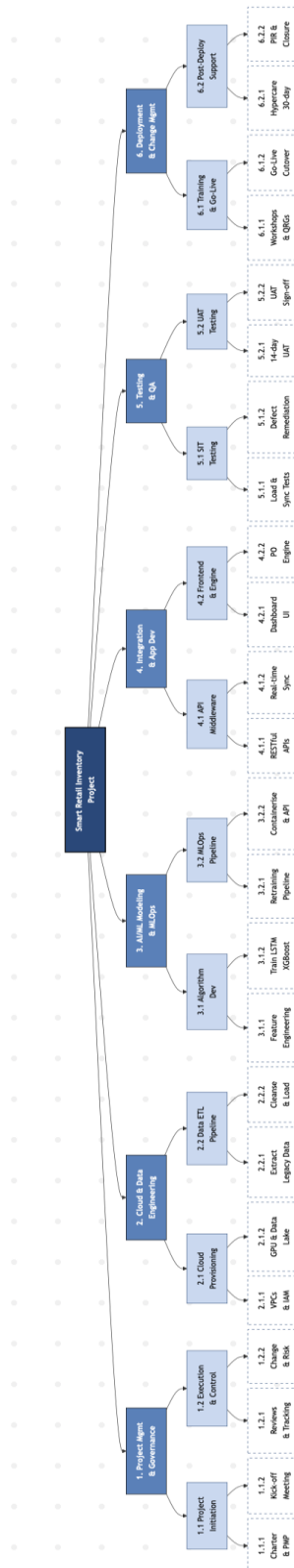
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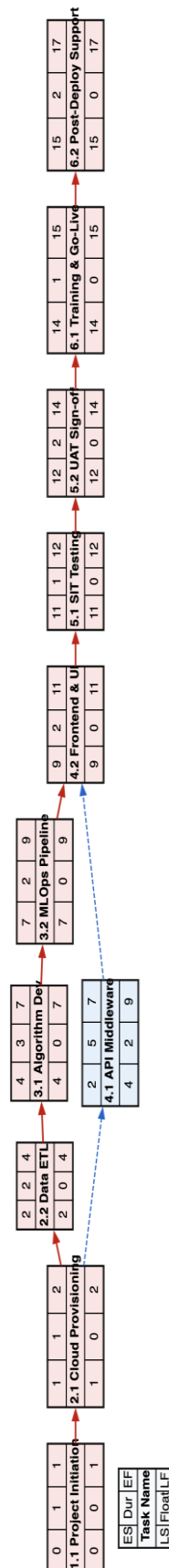
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Appendix

The Work Breakdown Structure



Network Diagram



Gantt Chart

